



AWS Global Networks for Transit Gateways User Guide

AWS Network Manager



AWS Network Manager: AWS Global Networks for Transit Gateways User Guide

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What is AWS Global Networks for Transit Gateways?

AWS Global Networks for Transit Gateways enables you to create one or more global networks and then centrally manage those global networks across AWS accounts, Regions, and on-premises locations.

Note

If you want to create a core network within one of your global networks you'll use AWS Cloud WAN to create, manage, and monitor that core network. For more information on creating a global network with a core network, see the [AWS Cloud WAN User Guide](#).

Global networks concepts

The following are the key concepts when using global networks to manage transit gateways.

- **Global network** — A single, private network that acts as the high-level container for your network objects. A global network can contain both AWS Transit Gateways and, if you're using AWS Cloud WAN, other Cloud WAN core networks. You can see these through the AWS Network Manager console.
- **Device** — Represents a physical or a virtual appliance in an on-premises network, data center, AWS Cloud, or other cloud providers.
- **Connection** — Represents connectivity between two devices. The connection can be between a physical or virtual appliance and a third-party virtual appliance inside a VPC, or it can be between physical appliances in an on-premises network.
- **Link** — Represents a single internet connection from a site.
- **Site** — Represents a physical on-premises location. It could be a branch, office, store, campus, or a data center.

Home Region

The home Region is the AWS Region where data related to your use of your AWS Global Networks for Transit Gateways global network is aggregated and stored. Global networks aggregates and stores this information in the home Region to provide you with a central dashboard with visualized

insights into your global network. Currently, global networks only supports US West (Oregon) as the home Region.

Important

- Global networks aggregates and stores regional usage data associated with the transit gateways specified from the AWS Regions you're using to the US West (Oregon) Region.
- Global networks aggregates and stores regional usage data associated with the transit gateways from the AWS GovCloud (US) Regions to the AWS GovCloud (US-West) Region.
- Once established, you can't change the home Region.

We aggregate and store this regional usage data from the AWS Regions you are using to US West (Oregon) using Amazon Simple Queue Service (SQS) and Amazon Simple Storage Service (S3). This data includes but is not limited to:

- Topology data for registered transit gateways
- Event data for transit gateways and VPNs
- Transit gateway IDs for registering transit gateways into a global network
- (Optional) Location data related to your device and site registrations
- (Optional) Provider and link data related to your link registrations
- (Optional) IP address and CIDR ranges used in transit gateway Connect peers

All movement and data aggregation occurs over a secure and encrypted channel and stored with encryption at rest.

AWS Global Networks for Transit Gateways uses Amazon Location Service to create maps of your global network. For more information about Amazon Location Service, see [Amazon Location Service](#).

AWS PrivateLink support

Network Manager supports AWS PrivateLink to create private connectivity between Network Manager and your VPCs. Using PrivateLink, you can establish secure and private connectivity without the need for using an internet gateway or any NAT devices to communicate with your VPCs.

While there is no price charge for using Network Manager, there is a price charge for using PrivateLink. See for more information.

 Note

- PrivateLink only supports IPv6 dual-stack endpoints.
- Support for PrivateLink through Network Manager is currently available only in the us-west-2 and us-gov-west-1 Regions.

For more information on PrivateLink, see the [AWS PrivateLink Guide](#).

IPv6 support

Network Manager supports Internet Protocol version 6 (IPv6) on dual-stack endpoints. Backwards compatibility is supported for IPv4 endpoints. For example, `networkmanager.us-west-2.api.aws`.

Region availability

AWS Global Networks for Transit Gateways is available in the following AWS Regions:

AWS Region	Description
af-south-1	Africa (Cape Town)
ap-east-1	Asia Pacific (Hong Kong)
ap-northeast-1	Asia Pacific (Tokyo)
ap-northeast-2	Asia Pacific (Seoul)
ap-northeast-3	Asia Pacific (Osaka)
ap-south-1	Asia Pacific (Mumbai)
ap-south-2	Asia Pacific (Hyderabad)
ap-southeast-1	Asia Pacific (Singapore)

AWS Region	Description
ap-southeast-2	Asia Pacific (Sydney)
ap-southeast-3	Asia Pacific (Jakarta)
ap-southeast-4	Asia Pacific (Melbourne)
ap-southeast-5	Asia Pacific (Malaysia)
ca-central-1	Canada (Central)
ca-west-1	Canada West (Calgary)
eu-central-1	Europe (Frankfurt)
eu-central-2	Europe (Zurich)
eu-north-1	Europe (Stockholm)
eu-south-1	Europe (Milan)
eu-south-2	Europe (Spain)
eu-west-1	Europe (Ireland)
eu-west-2	Europe (London)
eu-west-3	Europe (Paris)
il-central-1	Israel (Tel Aviv)
me-central-1	Middle East (UAE)
me-south-1	Middle East (Bahrain)
sa-east-1	South America (São Paulo)
us-east-1	US East (N. Virginia)
us-east-2	US East (Ohio)

AWS Region	Description
us-west-1	US West (N. California)
us-west-2	US West (Oregon)
us-gov-east-1	AWS GovCloud (US-East)
us-gov-west-1	AWS GovCloud (US-West)

How to get started with global networks for transit gateways

Use the following resources to help you use global networks:

- [How AWS Global Networks for Transit Gateways works](#)
- [Get started](#)
- [the section called “Access transit gateway network dashboards”](#)

Pricing

There are no additional fees for using global networks to manage transit gateways networks. You are charged the standard fees for the network resources that you manage in your global network (such as transit gateways). For more information about pricing, see [AWS Transit Gateway pricing](#).

How AWS Global Networks for Transit Gateways works

To use global networks for transit gateways, you first create a *global network* to represent your network. Initially, the global network is empty. You then register your existing transit gateways and define your on-premises resources in the global network. This enables you to visualize and monitor your AWS resources and your on-premises networks through the global networks dashboard on the AWS Network Manager console.

After you create your global network, you can monitor your networks through this dashboard. You can view network activity and health using Amazon CloudWatch metrics and Amazon CloudWatch Events. The global networks dashboard can help you identify whether issues in your network are caused by AWS resources, your on-premises resources, or the connections between them.

global networks does not create, modify, or delete your transit gateways and their attachments. To work with transit gateways, use the Amazon VPC console and the Amazon EC2 APIs.

Contents

- [Register transit gateways](#)
- [Define and associate your on-premises network](#)
- [Supported resource types](#)

Register transit gateways

You can register transit gateways that are in the same AWS account as your global network. When you register a transit gateway, the following transit gateway attachments are automatically included in your global network:

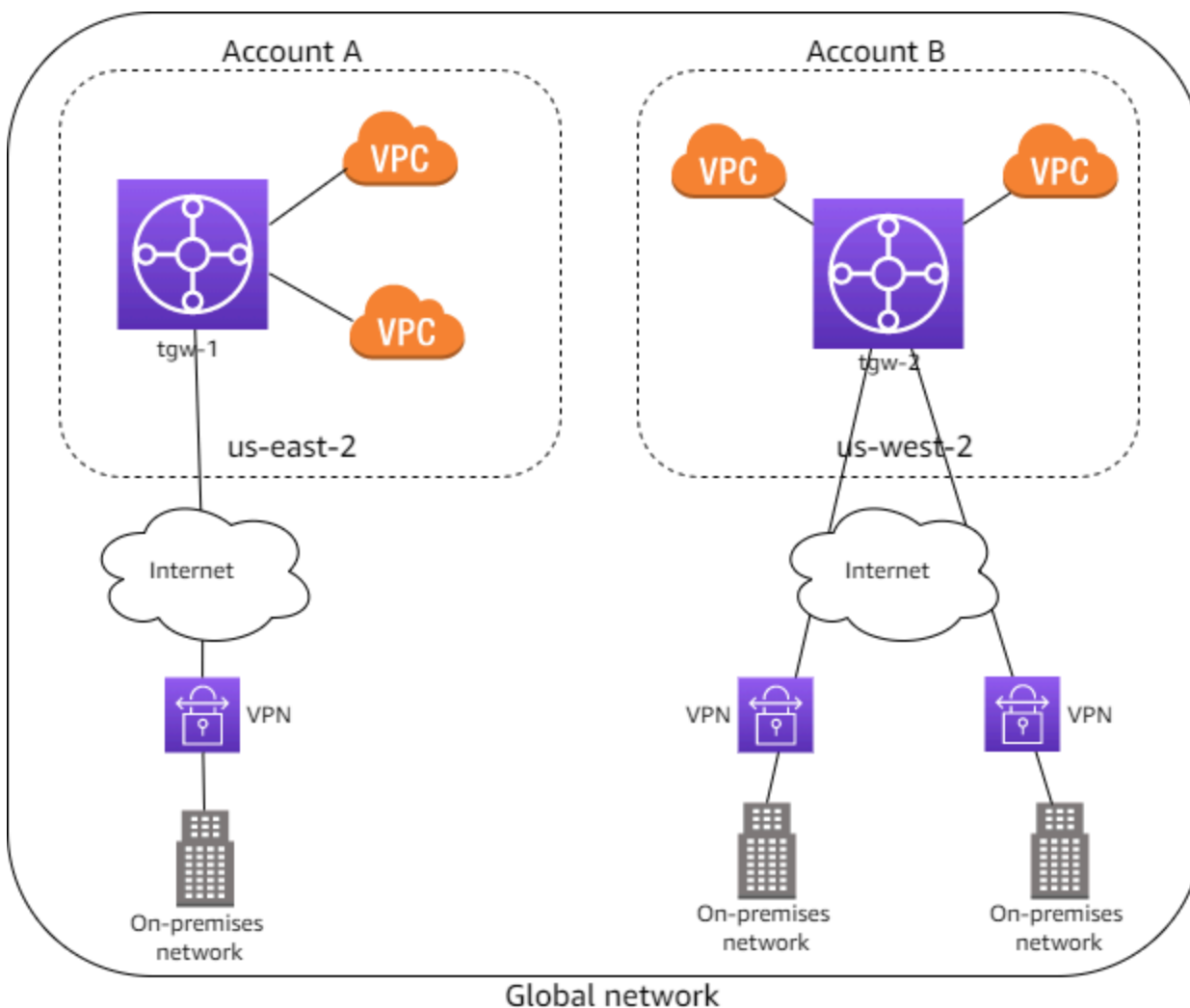
- VPCs
- Site-to-Site VPN connections
- AWS Direct Connect gateways
- Transit Gateway Connect
- Transit gateway peering connections

When you register a transit gateway that has a peering attachment, you can view the peer transit gateway in your global network, but you cannot view its attachments. If you own the peer transit gateway, you can register it in your global network to view its attachments.

If you delete a transit gateway, it's automatically deregistered from your global network.

Multi-Region and multi-account network

You can create a global network that includes transit gateways in multiple AWS Regions and accounts. This enables you to monitor the global health of your AWS network. In the following diagram, the global network includes a transit gateway in the us-east-2 Region from Account A and a transit gateway in the us-west-2 Region from Account B. Each transit gateway has VPC and VPN attachments. You can use the Network Manager console to view and monitor both of the transit gateways and their attachments.



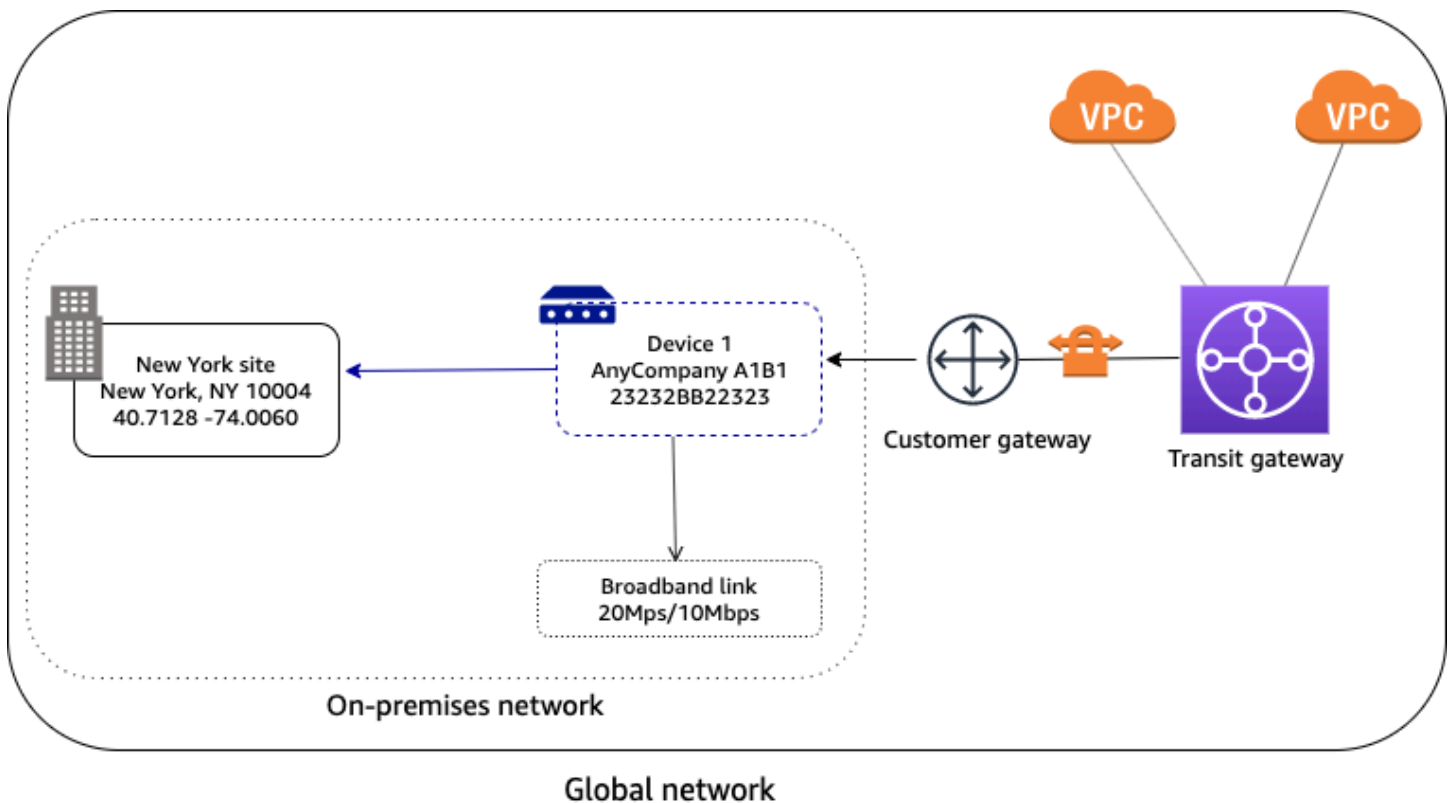
Define and associate your on-premises network

To represent your on-premises network, you add *devices*, *links*, and *sites* to your global network. A site represents the physical location of your branch, office, store, campus, or data center. When you add a site, you can specify the location information, including the physical address and coordinates.

A device represents the physical or virtual appliance that establishes connectivity with a transit gateway over an IPsec tunnel. A link represents a single outbound internet connection used by a device, for example, a 20-Mbps broadband link.

When you create a device, you can specify its physical location, and the site where it's located. A device can have a more specific location than the site, for example, a building in a campus or a floor in a building. When you create a link, you create it for a specific site. You can then associate a device with a link.

To connect your on-premises network to your AWS resources, associate a customer gateway that's in your global network with the device. If you've created a device to represent a virtual appliance sitting inside your VPC, and you've established a Connect peer from your virtual appliance to your AWS Transit Gateway, associate a Connect peer with the device to connect your virtual appliance network to your AWS resources. In the following diagram, the on-premises network is connected to a transit gateway through a Site-to-Site VPN connection.



You can have multiple devices in a site, which you can associate a device with multiple links. For examples, see [AWS Global Networks for Transit Gateways scenarios](#).

You can work with one of our Partners in the AWS Partner Network (APN) to provision and connect your on-premises networks. For more information, see [AWS Network Manager](#).

Supported resource types

After you register a transit gateway, you can view and monitor the resources in your global network.

Amazon VPC resources

Resource	Related resources
Transit gateway	- Transit gateway attachment - Transit gateway route table
Transit gateway attachment	- Direct Connect gateway - Transit gateway - Transit gateway attachment - Transit Gateway Connect peer - VPC - VPN connection
Transit gateway route table	Transit gateway
Transit Gateway Connect peer	- Device - Transit gateway attachment

AWS VPN resources

Resource	Related resources
Customer gateway	- Device - VPN connection
VPN connection	Customer gateway

- Transit gateway attachment

AWS Direct Connect resources

Resource

Direct Connect connection

Direct Connect gateway

Virtual interface

Related resources

- Virtual interface
- Transit gateway attachment
- Virtual interface
- Direct Connect connection
- Direct Connect gateway

AWS Network Manager resources

Resource

Connection

Device

Link

Site

Related resources

- Device
- Connection
- Customer gateway
- Link
- Site
- Transit Gateway Connect peer
- Device
- Site
- Device
- Link

Get started with AWS Global Networks for Transit Gateways

The following tasks help you become familiar with AWS Global Networks for Transit Gateways. For more information about how AWS Global Networks for Transit Gateways works, see [How global networks work](#).

In this example, you create a global network and register your transit gateway with the global network. You can also define and associate your on-premises network resources with the global network.

Tasks

- [Prerequisites](#)
- [Step 1: Create a global network](#)
- [Step 2: Register your transit gateway](#)
- [Step 3: \(Optional\) Define and associate your on-premises network resources](#)
- [Step 4: \(Optional\) Enable multi-account access](#)
- [Step 5: View and monitor your global network](#)

Prerequisites

Before you begin, ensure that you have a transit gateway with attachments in your account or in any account within your organization. For more information, see [Getting Started with Transit Gateways](#).

The transit gateway can be in the same AWS account as the global network or in a different AWS account within the organization.

Step 1: Create a global network

Create a global network as a container for your transit gateway.

To create a global network

1. Open the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.

2. Under **Connectivity**, choose **Global Networks**.
3. Choose **Create global network**.
4. Enter a name and description for the global network, and choose **Create global network**.

Step 2: Register your transit gateway

Register a transit gateway in your global network.

To register the transit gateway

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose **Register transit gateway**.
5. From the **Select account** dropdown list, choose the account that you want to register the transit gateway from.

A list of transit gateways from that account appear in the **Select transit gateway to register** section.

6. Select one or more transit gateways from the list, and then choose **Register transit gateway**.

Step 3: (Optional) Define and associate your on-premises network resources

You can define your on-premises network by creating sites, links, and devices to represent objects in your network. For more information, see the following procedures:

- [Create a site using AWS Network Manager](#)
- [Adding a link](#)
- [Add a device using AWS Network Manager](#)

You associate the device with a specific site, and with one or more links. For more information, see [the section called "Associate or disassociate a device link"](#).

On your transit gateway you can

- Create a Site-to-Site VPN connection attachment. For more information, see [the section called “Customer gateway associations”](#).
- Create a transit gateway Connect attachment, and then associate the Connect peer with the device. For more information, see [the section called “Add a Connect peer association”](#).

You can also work with one of our Partners in the AWS Partner Network (APN) to provision and connect your on-premises network. For more information, see [AWS Network Manager](#).

Step 4: (Optional) Enable multi-account access

Enable multi-account access to register transit gateways from multiple accounts, allowing you to view and manage transit gateways and associated resources from those registered accounts in your global network. Onboarding to AWS Organizations is a prerequisite for enabling multi-account access for Network Manager.

1. Create your organization using AWS Organizations.

If you've already done this skip this step. For more information on creating an organization using AWS Organizations, see [Creating and managing an organization](#) in the *AWS Organizations User Guide*.

2. Enable multi-account on the Network Manager console.

This enables trusted access for Network Manager and allows for registering delegated administrators. For more information enabling trusted access and registering delegated administrators, see [Multi-account in AWS Global Networks for Transit Gateways](#).

3. Create your global network.

For more information on creating a global network, see [Create a global network using AWS Network Manager](#).

4. Register transit gateways.

With multi-account enabled, you can register transit gateways from multiple accounts to your global network. For more information about registering transit gateways, see [Transit gateway registrations in AWS Global Networks for Transit Gateways](#).

Step 5: View and monitor your global network

The Network Manager console provides a dashboard for you to view and monitor both your transit gateway network objects in your global network.

To access the dashboard for your global network

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. The **Overview** page provides an inventory of the objects in your global network for your transit gateway network. For more information about the pages in the dashboard, see [the section called "Access transit gateway network dashboards"](#).

AWS Global Networks for Transit Gateways scenarios

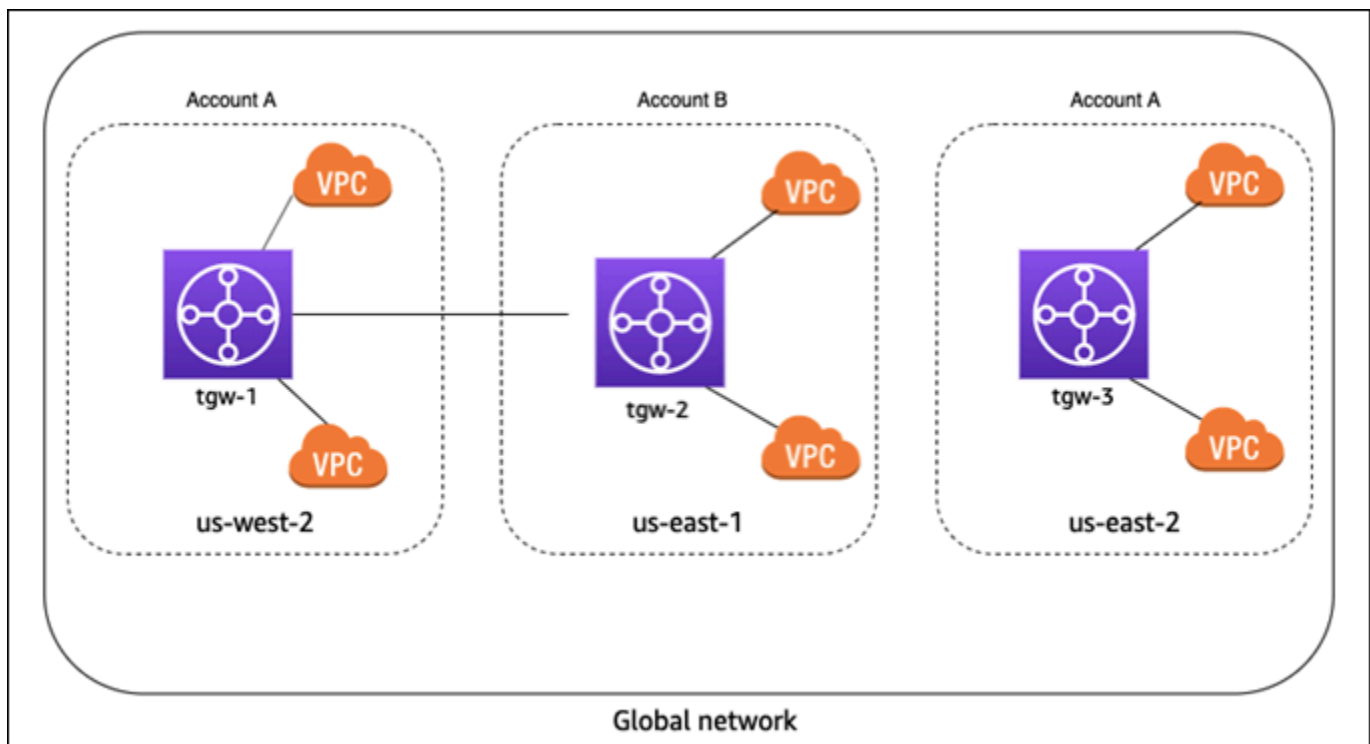
The following are common use cases and scenarios for using AWS Global Networks for Transit Gateways to manage your transit gateways.

Contents

- [AWS-only multi-Region and multi-account global network](#)
- [Single device with a single VPN connection](#)
- [Device with multiple VPN connections](#)
- [Multi-device and multi-link site](#)
- [SD-WAN connecting to AWS](#)
- [Connection between devices](#)

AWS-only multi-Region and multi-account global network

In this scenario, your AWS network consists of three transit gateways. You own transit gateways `tgw-1` and `tgw-3`. Transit gateway `tgw-1` has a peering attachment with transit gateway `tgw-2` that's in a different AWS account. Your entire network is within AWS, and does not consist of on-premises resources.



For this scenario, do the following in Network Manager:

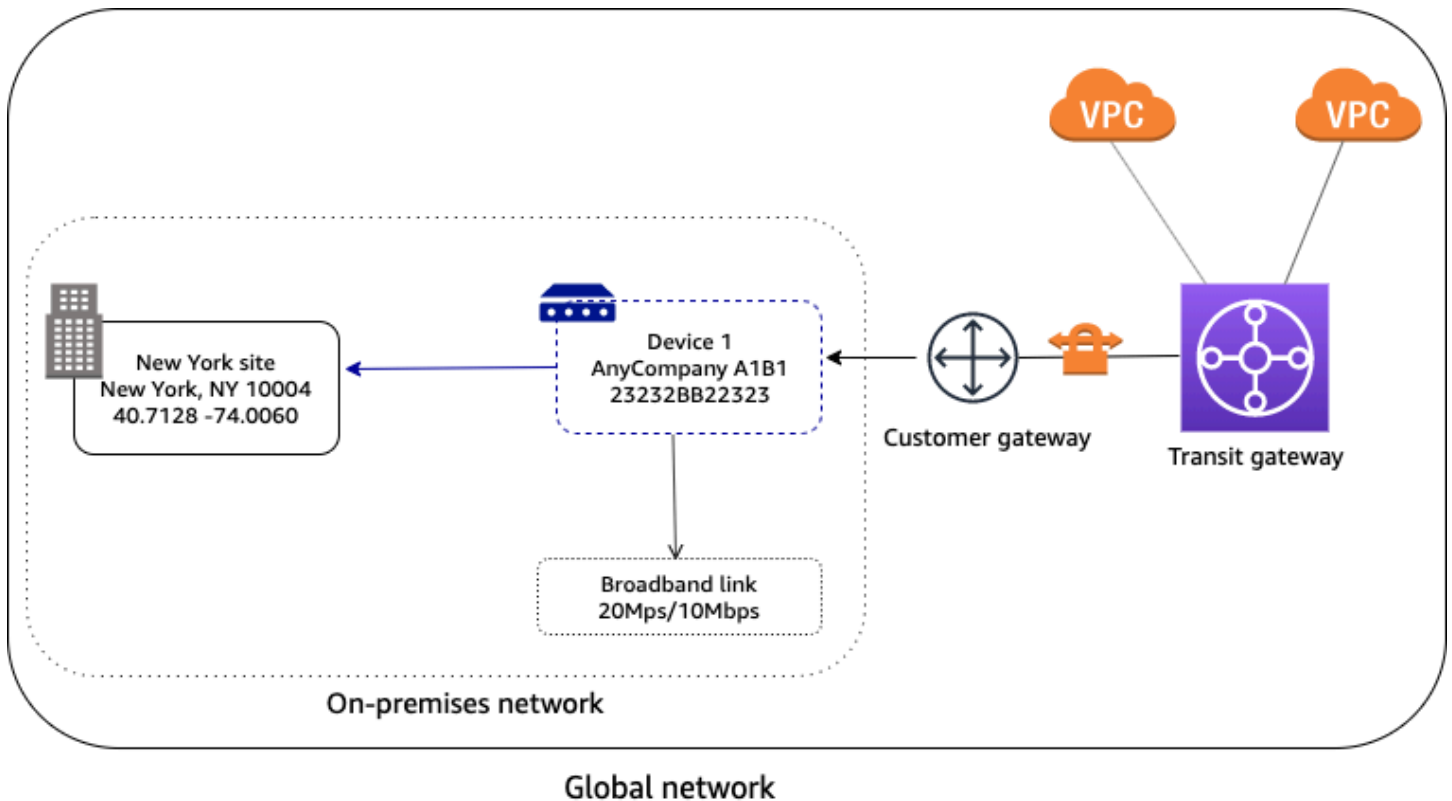
- Create a global network. For more information, see [Create a global network using AWS Network Manager](#).
- Register the transit gateways tgw-1 and tgw-3 with your global network. For more information, see [Register a transit gateway using AWS Network Manager](#).

When you register tgw-1, the transit gateway peering attachment is included in the global network, and you can see information about tgw-2. However, any attachments for tgw-2 are not included in your global network. To see attachments for tgw-2, you must enable multi-account access.

- This enables trusted access for global networks and allows for registering delegated administrators. For more information enabling trusted access and registering delegated administrators, see [Multi-account in AWS Global Networks for Transit Gateways](#).
- Register the tgw-2 transit gateway with your global network. For more information, see [Transit gateway registrations in AWS Global Networks for Transit Gateways](#).

Single device with a single VPN connection

In the following scenario, your global network consists of a single site with a single device and link. The site is connected to your AWS network through a Site-to-Site VPN attachment on a transit gateway. Your transit gateway also has two VPC attachments.



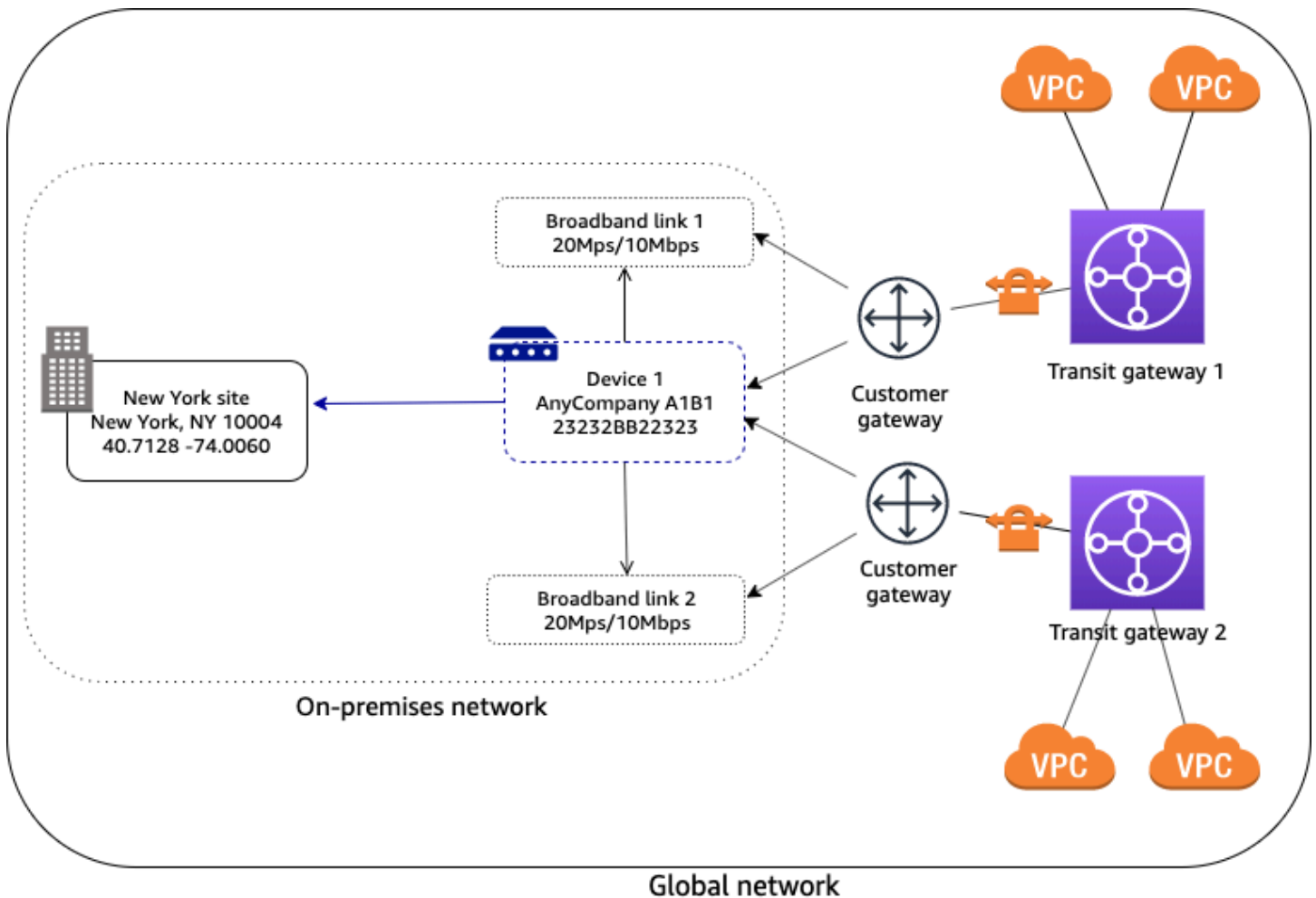
For this scenario, do the following in Network Manager:

- Create a global network. For more information, see [Create a global network using AWS Network Manager](#).
- Register the transit gateway. For more information, see [Register a transit gateway using AWS Network Manager](#).
- Create a site, device, and link. For more information, see [the section called "Sites and links"](#) and [the section called "Devices"](#).
- Associate the device with the site and with the link. For more information, see [the section called "Associate or disassociate a device link"](#).
- Associate the customer gateway (for the transit gateway Site-to-Site VPN attachment) with the device, and optionally, the link. For more information, see [the section called "Customer gateway associations"](#).

Device with multiple VPN connections

In the following scenario, your on-premises network consists of a device with two Site-to-Site VPN connections to AWS. The device is associated with two customer gateways on two different transit

gateways. Each VPN connection uses a separate link. To indicate which link applies to which VPN connection, you associate the customer gateway with both the device and the corresponding link.



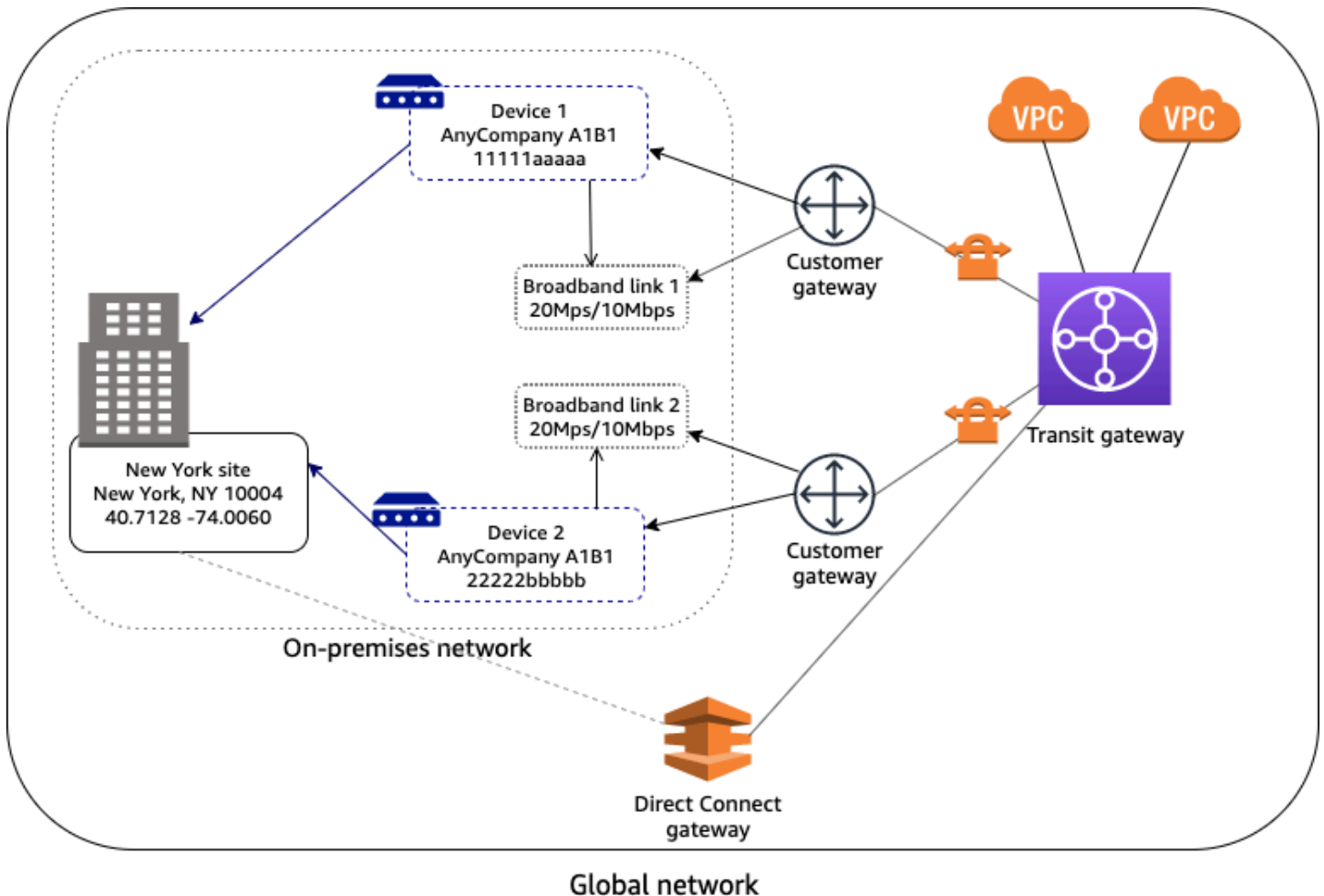
For this scenario, do the following in global networks:

- Create a global network. For more information, see [Create a global network using AWS Network Manager](#).
- Register the transit gateways. For more information, see [Register a transit gateway using AWS Network Manager](#).
- Create a site, device, and link. For more information, see [the section called “Sites and links”](#) and [the section called “Devices”](#). />.
- Associate the device with the site and both links. For more information, see [the section called “Associate or disassociate a device link”](#).
- Associate each customer gateway with the device and the corresponding link. For more information, see [the section called “Customer gateway associations”](#).

Multi-device and multi-link site

In the following scenario, your on-premises network consists of a site with two devices and two separate Site-to-Site VPN connections to AWS. For example, in a single building or campus, you might have multiple devices connected to AWS resources. Each device is associated with a customer gateway that's attached to your transit gateway.

Your AWS network is also connected to your on-premises network through an AWS Direct Connect gateway, which is an attachment on your transit gateway.



For this scenario, do the following in global networks:

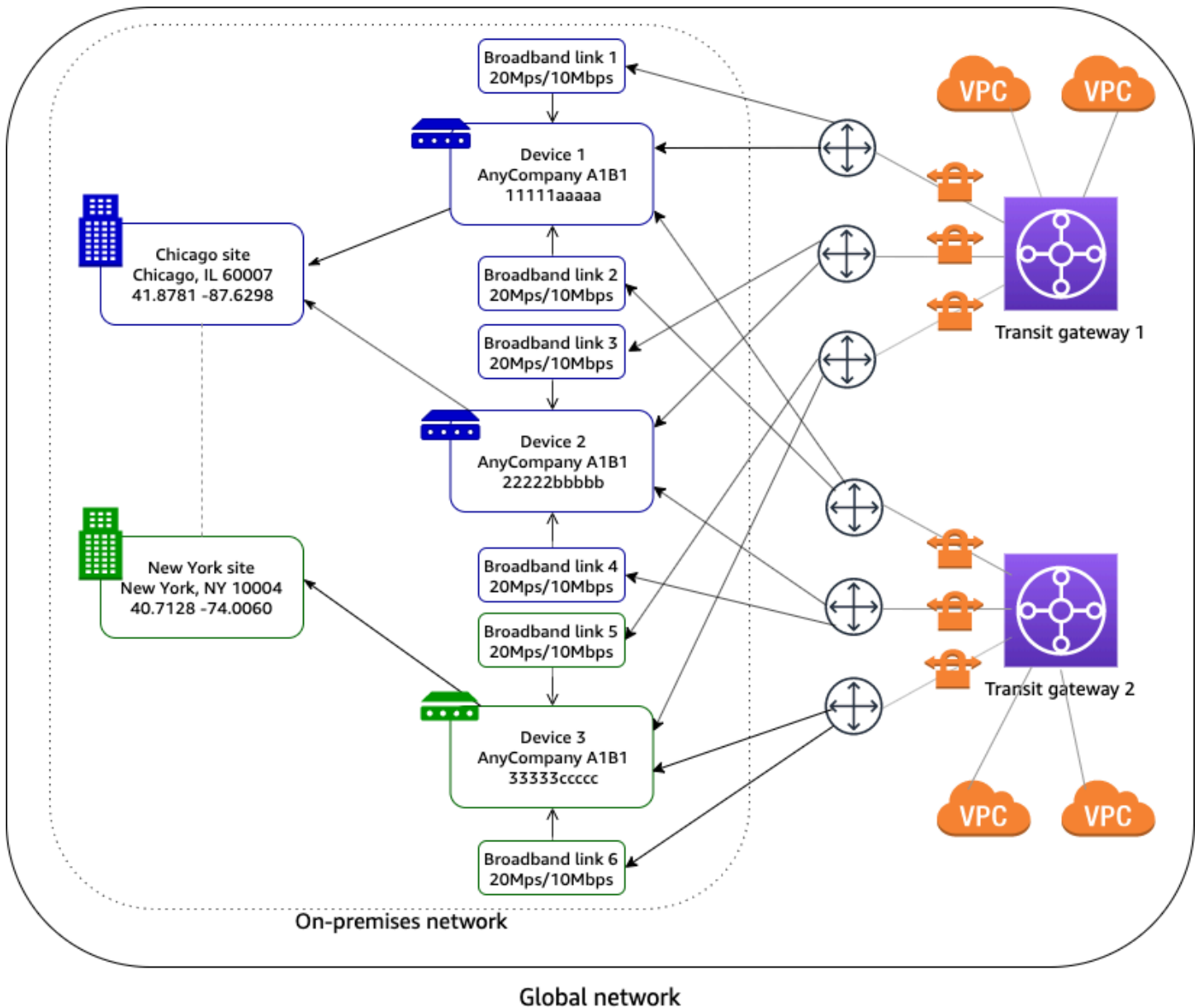
- Create a global network. For more information, see [Create a global network using AWS Network Manager](#).
- Register the transit gateway. For more information, see [Register a transit gateway using AWS Network Manager](#).

- Create one site, two devices, and two links. For more information, see [the section called “Sites and links”](#) and [the section called “Devices”](#).
- Associate each device with the corresponding link. For more information, see [the section called “Associate or disassociate a device link”](#).
- Associate each customer gateway with the corresponding device and link. For more information, see [the section called “Customer gateway associations”](#).

SD-WAN connecting to AWS

In the following example, your on-premises network consists of two sites. The Chicago site has two devices and the New York site has one device. Your AWS network consists of two transit gateways. All devices are associated with customer gateways (Site-to-Site VPN attachments) on both transit gateways.

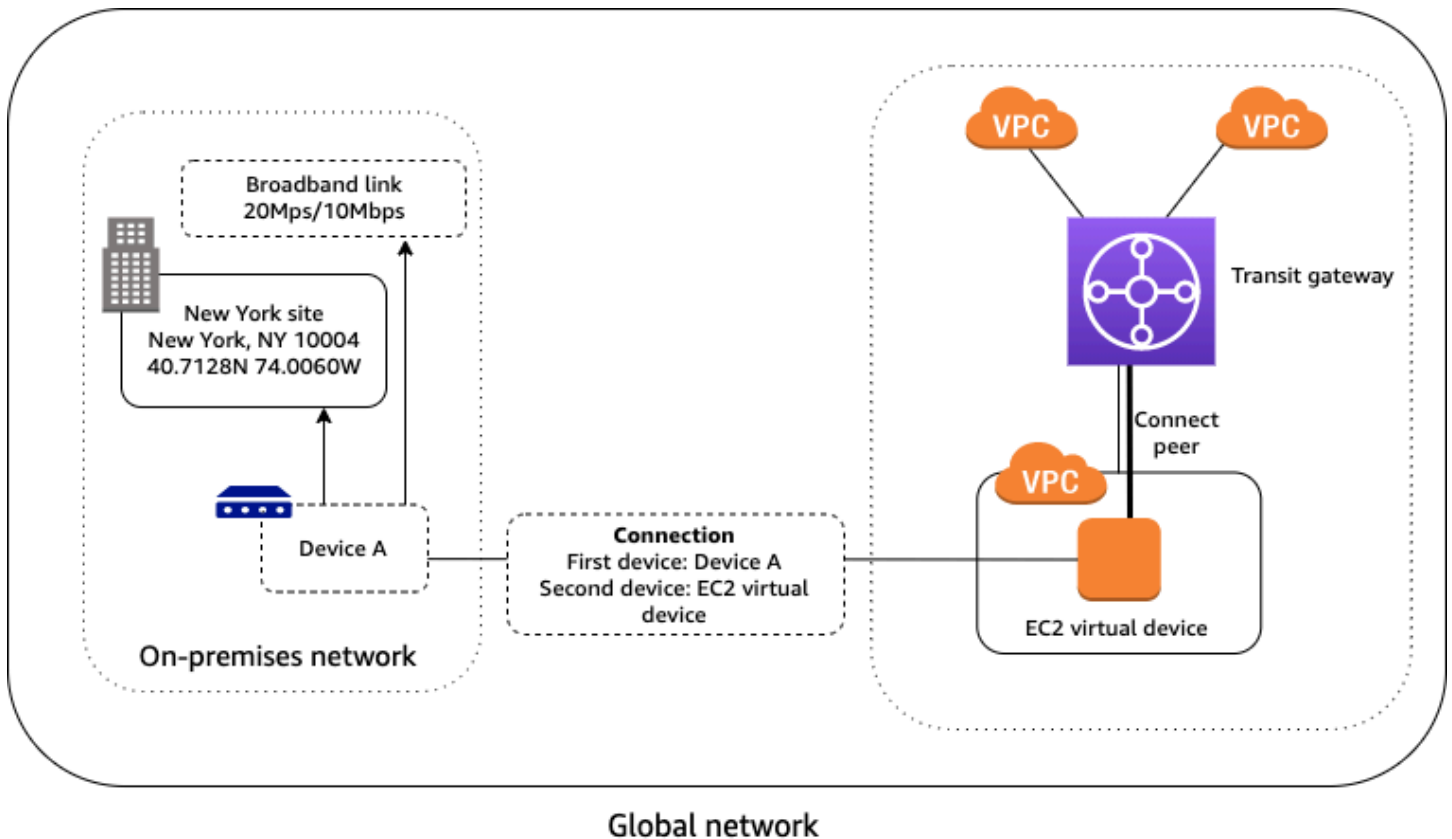
Your on-premises network is managed using SD-WAN. The SD-WAN controller creates Site-to-Site VPN connections to the transit gateways, and creates the device, site, and link resources in Network Manager. This automates connectivity and enables you to get a full view of your network in global networks. The SD-WAN controller can also use global networks events and metrics to enhance its dashboard.



For more information about Partners who can help you set up your Site-to-Site VPN connections, see [AWS Network Manager](#).

Connection between devices

In the following scenario, your AWS network consists of a transit gateway with a [Connect attachment](#) to a VPC that contains a virtual appliance on an EC2 instance. A Connect peer (GRE tunnel) is established between the transit gateway and the appliance. The appliance is connected to a physical device in your on-premises network through a connection.



For this scenario, do the following in global networks:

- Create a global network. For more information, see [Create a global network using AWS Network Manager](#).
- Register the transit gateway. For more information, see [Register a transit gateway using AWS Network Manager](#).
- Create a site, device, and link for your on-premises network. For more information, see [the section called "Sites and links"](#) and [the section called "Devices"](#).
- Associate the device with the site and with the link. For more information, see [the section called "Associate or disassociate a device link"](#).
- Create a device for the EC2 virtual device. For visualization in the global networks console, specify the AWS location of the device (for example, the Availability Zone). For more information, see [the section called "Devices"](#).
- Create a connection between the on-premises device and the virtual device. For more information, see [the section called "Associate or disassociate an on-premises link"](#).
- Associate the Connect peer with the on-premises device. For more information, see [the section called "Associate or disassociate a Connect peer"](#).

Modify a global network using AWS Network Manager

After setting up a global network you further modify it by performing a number of tasks.

Tasks you can perform to enhance your global network include:

- **Monitor and manage global network resources from other AWS accounts.**

AWS Global Networks for Transit Gateways supports multi-account access. If enabled, multi-account allows you to manage, monitor, and view dashboards of resources from AWS accounts shared with your account.

- **Register a transit gateway**

Register existing transit gateways in your global network. When you register a transit gateway all transit gateway attachments are automatically included in the registration. A transit gateway must first be created before it can be registered. For more information about transit gateways and creating one, see [Transit gateways](#) in the *Amazon VPC Transit Gateways User Guide*.

- **Create sites and links, and connect devices**

Add representations of physical devices and sites to your global network. You can then create a link that associates a device and a site.

- **Create customer gateway associations**

Create associations between two devices or between a device and a transit gateway Connect peer.

- **Access global network dashboards**

AWS Global Networks for Transit Gateways includes separate transit gateway network and transit gateway dashboards. On these dashboards you can view logical trees and geographic maps of your networks, which includes attachments, sites and devices. You can also view monitoring and events dashboards, allowing you to view Amazon CloudWatch metrics and to set threshold alarms on these metrics.

If your account is set up for multi-account, you can manage global network resources from multiple AWS accounts. For more information on multi-account, see [the section called “Multi-account”](#).

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- [Global networks in AWS Global Networks for Transit Gateways](#)
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- [Resource tags in AWS Global Networks for Transit Gateways](#)

Multi-account in AWS Global Networks for Transit Gateways

With AWS Global Networks for Transit Gateways, you can manage, monitor, and view dashboards of global network resources from multiple AWS accounts associated with a single organization using AWS Network Manager. For more information about setting up multi-account, see [Manage multiple accounts in global networks using AWS Organizations](#) below.

Important

- We strongly recommended that you use the global networks console for enabling multi-account settings with global networks, because the console automatically creates all required roles and permissions for multi-account access. Choosing an alternative approach requires an advanced level of expertise, and opens the multi-account set up for your global network to be more prone to error.
- Multi-account is not available in the AWS GovCloud (US-West) and the AWS GovCloud (US-East) Regions.

Prerequisites

To enable multi-account, you first set up an account in AWS Organizations. This first account becomes the management account. Using this account, you can then add other accounts as member accounts to your organization. For more information about how multi-account support works, see [Creating and managing an organization](#) in the *AWS Organizations User Guide*.

Manage multiple accounts in global networks using AWS Organizations

AWS Global Networks for Transit Gateways allows you to centrally manage, monitor, and visualize network resources from multiple accounts within an organization in a single global network. To manage resources from multiple accounts in global networks, you first set up an organization using AWS Organizations. The first account that you use to create an organization becomes the management account. Using this account, you can add other accounts as member accounts to your organization. From the management account, you can designate one or more accounts within the organization as delegated administrator accounts by registering them using the global networks console. For more information about setting up an organization, see [Creating and managing an organization](#) in the *AWS Organizations User Guide*.

To enable multi-account access in the global networks console, you first enable trusted access for the Network Manager service, and then register a delegated administrator account for your organization.

Important

- We strongly recommended that you use the global networks console for enabling multi-account settings with global networks, because the console automatically creates all required roles and permissions for multi-account access. Choosing an alternative approach requires an advanced level of expertise, and opens the multi-account set up for your global network to be more prone to error.
- Multi-account is not available in the AWS GovCloud (US-West) and the AWS GovCloud (US-East) Regions.

With multi-account support, you can create a single global network for any of your AWS accounts, and then register transit gateways from those accounts using the global networks console. Multi-account is supported in all AWS Regions where global networks is supported. For more information about multi-account, see [Multi-account in AWS Global Networks for Transit Gateways](#).

Trusted access

Trusted access creates `AWSServiceAccess` for global networks and AWS CloudFormation StackSets with AWS Organizations. Enabling trusted access provides required permissions for AWS Organizations to deploy service-linked roles (SLRs) to all member accounts within your organization.

Enable trusted access

When you enable trusted access from the global networks console, you select a one-time permission level (`IAMRoleForAWSNetworkManagerCrossAccountResourceAccess`) as either administrator or read-only for each of the management and delegated administrator accounts.

- **Admin** — Assign this permission if the delegated administrator and management accounts need to be able to modify resources from other accounts in the global network while using the global networks console switch role.
- **Read-only** — Assign this permission if the delegated administrator and management accounts only need to review information about resources from other accounts in the global network while using the global networks console switch role, but don't need to make any changes.

The global networks console manages all of this when calling the Network Manager API.

When you enable trusted access, the following roles are deployed in your organization using AWS CloudFormation StackSets and AWS Identity and Access Management (IAM) services:

- The Network Manager SLR (`AWSServiceRoleForNetworkManager`) to all member accounts
- The AWS CloudFormation StackSets member SLR (`AWSServiceRoleForCloudFormationStackSetsOrgMember`) to all member accounts
- The Network Manager SLR (`AWSServiceRoleForNetworkManager`) to the management account
- The AWS CloudFormation StackSets admin (`AWSServiceRoleForCloudFormationStackSetsOrgAdmin`) SLR to the management account
- The Amazon CloudWatch sharing role (`CloudWatch-CrossAccountSharingRole`) to all member accounts
- The global networks console switch role (`IAMRoleForAWSNetworkManagerCrossAccountResourceAccess`) to all member accounts
- The Amazon CloudWatch monitoring role (`AWSServiceRoleForCloudWatchCrossAccount`) to the management account

For more information about enabling trusted access, see [Enable trusted access in an AWS global network](#).

Disable trusted access

Note

Disabling trusted access through the global networks console removes `AWSServiceAccess` for global networks with AWS Organizations. Disabling trusted access removes global networks access to perform tasks within your organization. AWS Organizations won't allow you to disable an organization's trusted access for the Network Manager service if there are any delegated administrators that haven't been deregistered from that organization.

- Disabling trusted access through the global networks console won't remove `AWSServiceAccess` for AWS CloudFormation StackSets with AWS Organizations. You can manually remove the service access for AWS CloudFormation StackSets by using the AWS CloudFormation StackSet console or by using the Organizations API/CLI. For more information on disabling trusted access for AWS CloudFormation StackSets, see [Disable trusted access with AWS CloudFormation StackSets](#) in the *AWS Organizations User Guide*.
- Disabling trusted access won't remove any SLRs that were deployed when enabling trusted access.

When you disable trusted access, the following are affected in global networks:

- All transit gateways owned by other accounts in your organization. You won't be able to see transit gateways or their attached resources from other accounts in your organization that were registered to your global network.
- IAM roles deployed in all member accounts managed by the Network Manager service. Disabling trusted access doesn't remove accounts, transit gateways, or resources but does deregister them from other delegated administrator's global networks. These can be added back in as needed by re-enabling trusted access. For more information about the `DeleteStackSet` API, see [DeleteStackSet](#) in the *AWS CloudFormation API Reference*.

For more information about disabling trusted access, see [Disable trusted access in an AWS global network](#).

Delegated administrators

Member accounts in your organization with delegated administrator access are able to leverage service-linked roles and assume IAM roles for access across multiple accounts. Only member accounts that are part of your AWS Organizations can be registered as delegated administrators. Your organization can have up to ten registered delegated administrators. Before you register a delegated administrator, you must enable global networks trusted access for your organization. For more information, see [Enable trusted access in an AWS global network](#).

Important

Using your AWS Organizations management account to manage your global network in global networks is not recommended because the required service-linked roles are not propagated to this account. For more information on service-linked roles, see [the section called "Service-linked role"](#).

Register delegated administrators

After it's registered, a delegated administrator has the same permissions as the management account. A delegated administrator for the Network Manager service can leverage the SLRs in the member accounts that were deployed when trusted access was enabled and can view transit gateways from other member accounts and can register them to your global network. This allows transit gateways and associated resources to appear in your global network topology. In addition AWS CloudFormation StackSets is updated to include the delegated administrator accounts in the trusted relationship of the deployed IAM roles in the member accounts.

For information about registering a delegated administrator, see [Register an administrator for multi-account in an AWS global network](#).

Deregister delegated administrators

Deregistering a delegated administrator removes that account's permission to leverage SLRs and assume IAM roles in other member accounts that were set up using AWS Organizations.

After it's deregistered, the delegated administrator no longer has the same permissions as the management account. The following occurs:

- A delegated administrator is no longer able to leverage the deployed SLRs in the member accounts that were deployed when trusted access was enabled.

- All registered transit gateways from other member accounts are deregistered from any global network for the specific delegated administrator. The network topology is updated to no longer show resources from other member accounts.
- AWS CloudFormation StackSets are updated with the removal of the delegated administrator account. That account is no longer able to assume any IAM roles deployed in other member accounts.

For information about deregistering a delegated administrator, see [Deregister an administrator from multi-account in an AWS global network](#).

Multi-account tasks

- [Enable trusted access in an AWS global network](#)
- [Disable trusted access in an AWS global network](#)
- [Register an administrator for multi-account in an AWS global network](#)
- [Deregister an administrator from multi-account in an AWS global network](#)
- [Manage IAM multi-account role deployments in an AWS global network](#)
- [Troubleshoot multi-account self-managed roles in an AWS global network](#)

Enable trusted access in an AWS global network

Enabling trust is a one-time task that deploys the required service-linked roles (SLRs) and custom Identity and Access Management (IAM) roles to all accounts in your organization that can be assumed by the management account or [delegated administrators](#) for access across multiple accounts. For more information about trusted access, see [Trusted access](#).

To enable multi-account trusted access

1. Log into the global networks console at <https://console.aws.amazon.com/networkmanager/home/>, using the AWS Organizations management account.
2. Choose **Get started**.
3. In the navigation pane, choose **Enable trusted access**.
4. From the **Permission level** dropdown list in **Enable trusted access**, choose the Permission level for the Network Manager console switch role `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess`. This role is deployed to all member accounts and is assumed by the delegated administrator or management account

when accessing resources from other accounts using the global networks console. You can choose only one permission level for all accounts. Permission can be one of the following:

- **Read-only** — Assign this permission if the delegated administrator and management accounts only need to review information about resources from other accounts in the global network while using the console switch role, but don't need to make any changes.
- **Admin** — Assign this permission if the delegated administrator and management accounts need to be able to modify resources from other accounts in the global network while using the global networks console switch role.

5. Choose **Enable trusted access**.

Depending on your organization size, it might take a few minutes or more to enable trusted access. During this time the **State** shown in the **Trusted access** section displays **Enabling in progress**. When access is enabled, the **State** changes to **Enabled**. Additionally, the **IAM role deployments status** section at the bottom of the page displays the status of the IAM roles being deployed to member accounts of the organization.

6. After trusted access is enabled, you can register delegated administrators.

Disable trusted access in an AWS global network

Disabling trusted access removes the trust relationship between the Network Manager service access and your organization. Network Manager is no longer able to perform actions within your organization or access information about your organization. Trusted access remains for AWS CloudFormation StackSets in the event that your organization is using that service outside of Network Manager. For more information on disabling AWS CloudFormation StackSets, see [Disabling trusted access with AWS CloudFormation Stacksets](#) in the *AWS Organizations User Guide*.

Transit gateways from other accounts are deregistered from global networks owned by the management account and can no longer provide access to their attached resources. For more information about disabling trusted access, see [Disable trusted access](#).

You must first deregister all delegated administrators before you can disable trusted access. If you have registered delegated administrators, you will be prompted to deregister them during the disable trusted access process.

You can enable trusted access again after disabling it. However you will need to set up the list of delegated administrators again.

To disable trusted access

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/> with the management account.
2. Under **Connectivity**, choose **Global Networks**.
3. In the navigation pane, choose **Settings**.
4. In the **Trusted Access** section, choose **Disable trusted access**.
5. If you have any registered delegated administrators, you can deregister them by choosing **Deregister delegated administrators**.
6. Choose **Disable trusted access** on the confirmation dialog box to confirm that you want to disable trusted access.

Depending on the size of your organization, it might take several minutes or longer to disable trusted access. The **State** displays **Disabling in progress**. During this time you won't be able to re-enable trusted access. When finished, the Status changes to **Disabled**.

Register an administrator for multi-account in an AWS global network

Use the AWS Global Networks for Transit Gateways console to register delegated administrators. You can register up to ten delegated administrators. Delegated administrators can assume the SLR and IAM roles deployed while enabling trusted access for access across multiple accounts. For more information about delegated administrators, see [Delegated administrators](#).

To register a delegated administrator

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/> with the management account.
2. Under **Connectivity**, choose **Global Networks**.
3. In the navigation pane, choose **Settings**.
4. In the **Delegated Administrators** section, choose **Register delegated administrator**.
5. From the **AWS account ID** dropdown list, choose one or more AWS Organizations accounts that you want to delegate administrator permissions to.
6. Choose **Register delegated administrator**.
7. When the delegated administrator is registered, you can then register transit gateways from any transit gateways from any account within your organization to the global network in the

delegated administrator account. For more information about registering transit gateways in the global network of a delegated administrator account, see [Transit gateway registrations in AWS Global Networks for Transit Gateways](#).

Deregister an administrator from multi-account in an AWS global network

Deregistering delegated administrators removes that account's permission to manage global networks for your organization. All registered transit gateways from other member accounts are deregistered from the specific delegated administrator's global networks. For more information about how deregistering delegated administrators works, see [Deregister delegated administrators](#).

To deregister a delegated administrator

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/> with the management account.
2. Under **Connectivity**, choose **Global Networks**.
3. In the navigation pane, choose **Settings**.
4. In the **Delegated Administrators** section, choose one or more accounts that you want to deregister.

Depending on your organization size and the number of delegated administrators you're deregistering, this could take several minutes. During this time you won't be able to register any new delegated administrators.

Manage IAM multi-account role deployments in an AWS global network

The **IAM role deployments status** section displays the current role deployments status for all member accounts set up in your account.

- **Member account ID** — The account ID for the account set up in AWS Organizations. This includes member accounts and members that have been registered as delegated administrators.
- **CloudWatch role status** — The status of the account's Amazon CloudWatch role. If you enable multi-account using the Network Manager console, this is **StackSets-managed** if deployed successfully. Otherwise, this is **Self-managed**.

- **Console role status** — The status of the account's Network Manager console role. If you enable multi-account using the Network Manager console, this is **StackSets-managed** if deployed successfully. Otherwise, this is **Self-managed**.
- **Review required** — This applies only to **Self-managed** roles. A review is required to ensure that the permissions set up for the account are correct. For more information, see [Multi-account access roles for AWS Global Networks for Transit Gateways](#).

If you make changes to your role policies, or if you've updated a self-managed role, you can deploy the updated policy to your AWS Organizations accounts.

To retry the IAM role deployment status

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/> with the management account.
2. Under **Connectivity**, choose **Global Networks**.
3. In the navigation pane, choose **Settings**.
4. In the **IAM role deployments status** section, choose **Retry role deployment**.

Depending on your organization size and the number of member accounts in your organization, this could take several minutes. During this time you won't be able to register or deregister any new delegated administrators.

Troubleshoot multi-account self-managed roles in an AWS global network

AWS Global Networks for Transit Gateways uses AWS CloudFormation StackSets to deploy the required `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess` role and the CloudWatch monitoring `CloudWatch-CrossAccountSharingRole` role in your AWS Organizations member accounts for cross-account access. For a CloudFormation StackSets-managed deployment, IAM roles must have the required policies attached, as well as the trusted relationship to allow registered delegated administrators and the management account the ability to assume these roles. In a self-managed deployment, you own the responsibility to attach the appropriate policies and to manage the trusted relationship required for the delegated administrator and management accounts to access multiple accounts.

Important

We strongly recommend that you use the global networks console for enabling multi-account settings using the global networks console as this automatically sets up all required roles and permissions for multi-account access. Choosing an alternative approach requires an advanced level of expertise and opens the multi-account setup for your global network to be more prone to error.

If the CloudFormation StackSets deployment fails, and the **Review required** message is **IAM role exists**, follow the steps below in [IAM role exists](#) to change the role from **Self-managed** to **StackSets-managed**. For any message other than **IAM role exists**, file an AWS Support case. For more information on creating a support case, see [Creating a support case](#) in the *AWS Support User Guide*.

IAM role exists

If the IAM role has the exact same name in a current the member account, these roles appear in the **IAM role deployments status** with a status of **Self-managed**. In order to change this to StackSets-managed, delete the IAM role from the member account with the duplicate role name. After deleting the IAM role, use the global networks console to retry the role deployment. For the steps to retry a role deployment, see [Manage IAM multi-account role deployments in an AWS global network](#) to retry the role deployment.

To change a role from self-managed to StackSets-managed

1. Access the AWS Identity and Access Management (IAM) console at <https://console.aws.amazon.com/iamv2/home?#/> with the member account that has a self-managed role status.
2. In the navigation pane, choose **Roles**.
3. In the **Roles** field, search for the role name you want to delete.
4. Choose the role, and then choose **Delete**.
5. Confirm that you want to delete the role.

 Warning

This might break other functionality if a custom role has other attached policies or trusted relationships.

6. Access the global networks console at <https://console.aws.amazon.com/networkmanager/home/> with the AWS Organizations management account.
7. Choose **Get started**.
8. In the navigation pane, choose **Settings**.
9. In the **IAM role deployment status section**, choose **Retry role deployment**.

Depending on the size of your organization, it might take several minutes or longer to disable trusted access. During this time you won't be able to re-enable trusted access.

Global networks in AWS Global Networks for Transit Gateways

A global network in AWS Global Networks for Transit Gateways is a container for your network objects. When you create a global network, you create only the framework of the global network itself. You'll then further define this network by adding network objects to it, such as

- transit gateways.
- sites, devices, and links, and then creating connections using links between those sites and devices.
- customer gateway associations.
- Connect peer associations.

 Note

When creating a global network, you're prompted to create an associated core network. A core network is a feature of AWS Cloud WAN and is not needed if you're not using this feature. While creating a global network, you're prompted whether or not to create a core network. Clear the option to create the core network. If you decide later on that you want to create a core network for this global network you can. For the steps to create a core network, see [What is AWS Cloud WAN](#) in the *AWS Cloud WAN User Guide*.

You can create, view, update, and delete a global network using either the AWS Network Manager console or by using the CLI.

Global network tasks

- [Create a global network using AWS Network Manager](#)
- [View a global network using AWS Global Networks for Transit Gateways](#)
- [Update a global network using AWS Network Manager](#)
- [Delete a global network using AWS Network Manager](#)

Create a global network using AWS Network Manager

Create a global network.

To create a global network

1. Open the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. Choose **Create global network**.
4. Enter a **Name** and **Description** for your global network.
5. (Optional) In Additional settings, add **Key** and **Value** tags that further help identify an Network Manager resource. To add multiple tags, choose **Add tag** for each tag you want to add.
6. Choose **Next**.
7. To create a AWS Transit Gateway network only, clear the **Add core network in your global network** check box on the **Create global network - optional** page, and then choose **Next**.

Note

Core networks are only used with AWS Cloud WAN. If you're creating global network for AWS Cloud WAN and want to create a core network, see [Create a core network policy](#) in the *AWS Cloud WAN User Guide*.

8. Review the information for the global network you want to create, and then choose **Create global network**.

To create a global network using the AWS CLI

Use the [create-global-network](#) command.

View a global network using AWS Global Networks for Transit Gateways

View the details of your global network and information about the network objects in your global network.

To view your global network information

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. The **Overview** page displays an inventory of the objects in both your core network and transit gateway network. To view details about the global network resource (such as its ARN), choose **Details**. For more information about the other pages on the dashboard, see [the section called "Access transit gateway network dashboards"](#).

To view global network details using the AWS CLI

Use the [describe-global-networks](#) command.

Update a global network using AWS Network Manager

Update a global network by modifying the description or tags.

To update your global network

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. Choose **Edit**.
5. For **Description**, enter a new description for the global network.

6. For **Tags**, choose **Remove tag** to remove an existing tag, or choose **Add tag** to add a new tag.
7. Choose **Edit global network**.

To update a global network using the AWS CLI

Use the [update-global-network](#) command to update the description. Use the [tag-resource](#) and [untag-resource](#) commands to update the tags.

Delete a global network using AWS Network Manager

Delete a global network framework. You cannot delete a global network if there are any network objects in the global network, including transit gateways, links, devices, and sites. You must first deregister or delete the network objects.

To delete your global network

1. Open the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. In the navigation pane, choose **Global networks**.
4. Choose your global network and choose **Delete**.
5. In the confirmation dialog box, choose **Delete**.

To delete a global network using the AWS CLI

Use the [delete-global-network](#) command.

Transit gateway registrations in AWS Global Networks for Transit Gateways

You can register your existing transit gateways with a global network. Any transit gateway attachments (such as VPCs, VPN connections, and AWS Direct Connect gateways) are automatically included in your global network.

Transit gateway limitations

Note the following about registering transit gateways in a global network:

- A transit gateway must first be created in Amazon Virtual Private Cloud (VPC) before it can be registered in a global network. For more information about transit gateways and creating one, see [Transit gateways](#) in the *Amazon VPC Transit Gateways User Guide*.
- You can have multiple global networks, but you can only register one transit gateway with one global network.
- You can register transit gateways that are in the same AWS account as the global network.
- You cannot create, delete, or modify your transit gateways and their attachments using the Network Manager console or APIs. To work with transit gateways, use the Amazon VPC console or the Amazon EC2 APIs.

Transit gateway registration tasks

- [Register a transit gateway using AWS Network Manager](#)
- [View registered transit gateways using AWS Network Manager](#)
- [Deregister a transit gateway using AWS Network Manager](#)

Register a transit gateway using AWS Network Manager

Register a transit gateway created using Amazon Virtual Private Cloud with your AWS global network using either the Network Manager console or using the CLI. You cannot register a transit gateway with more than one global network.

To register a transit gateway

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose **Register transit gateway**.
5. (Optional) If your account is enabled for multi-account access, from the **Select account** dropdown list choose the account you want to register transit gateways from.

The **Select transit gateway to register** section populates with that account's transit gateways.

6. Choose one or more transit gateways, and then choose **Register transit gateway**.

To register a transit gateway using the AWS CLI

Use the [register-transit-gateway](#) command.

View registered transit gateways using AWS Network Manager

View a transit gateway registered with your AWS global network using either the Network Manager console or using the CLI.

To access your registered transit gateways

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**.
5. The **Transit gateways** page lists your registered transit gateways. Choose the ID of transit gateway to view its details.

To view your registered transit gateways using the AWS CLI

Use the [get-transit-gateway-registrations](#) command.

Deregister a transit gateway using AWS Network Manager

Deregister a transit gateway from a global network using either the Network Manager console or using the CLI. Once deregistered, you can re-register this transit gateway with the same global network or with a different global network.

To deregister a transit gateway

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**.
5. Select your transit gateway, and choose **Deregister**.

To deregister a transit gateway using the AWS CLI

Use the [deregister-transit-gateway](#) command.

Sites and links in AWS Global Networks for Transit Gateways

After you've added any devices to your AWS global network, you can associate of your devices with that particular site using a connection, or link. For information on adding devices, see [the section called "Devices"](#).

Sites

A site represents the physical location of your network, using location information such as latitude, longitude, and address. You can have multiple sites for each of your network locations. Sites are useful when viewing the global network dashboard, which provides you the geographical location of these sites based on location information you provided. Once you create a site you can view the devices associated with the site and create links between devices and sites. You can also view any VPNs associated with the site as well as monitor CloudWatch metrics for this site.

Links

A link represents the connection between a device and a site. Once you've added a device and created a site, you can create an association between the device and a site.

Tasks

Create a site using AWS Network Manager

Create a site to represent the physical location of your network. Location information is used in the Network Manager transit gateway dashboards.

To create a site

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**. Choose **Create site**.

5. For **Name** and **Description**, enter a name and description for the site.
6. For **Address**, enter the physical address of the site, for example, New York, NY 10004.
7. For **Latitude**, enter the latitude coordinates for the site, for example, 40.7128.
8. For **Longitude**, enter the longitude coordinates for the site, for example, -74.0060.
9. Choose **Create site**.

Creating and viewing a site using the AWS CLI

Use the following commands:

- To create a site: [create-site](#)
- To view your sites: [get-sites](#)

View site details using AWS Network Manager

View details about a core network site using the Network Manager console.

To view details about a site

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**.
5. Choose the link that you want to see site details for.
6. The **General details** page provides information about the site.
7. Choose the **Devices** tab. This page displays information about the devices that are connected to the site. If you don't see a device listed, you'll need to add it. For more information on adding devices, see [the section called "Add a device"](#).
8. Choose the **Links** tab. This page displays the links that represent a connection from a device. If you don't see a link listed, you'll need to create the link. For the steps to create a link, see [the section called "Add a link"](#).
9. Choose the **VPNs** tab. This page displays site-related VPN information.

10. Choose the **Monitoring** tab. This page displays **Data In** and **Data Out** information for your links.
11. From the dropdown list, choose the link that you want to view information for.
12. (Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Update a site using AWS Network Manager

Update any of the details of a site you've added, including the description, address, latitude, and longitude.

To update a site

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**, and select your site.
5. Choose **Edit**.
6. Update the **Description**, **Address**, **Latitude**, **Longitude**, and **Tags** as needed.
7. Choose **Edit site**.

Update a site using the AWS CLI

Use the [update-site](#) command.

Delete a site using AWS Network Manager

Delete sites from your global network if the site is no longer valid or you no longer want to return any information about the site. You must first disassociate the site from any devices and delete any links for the site.

To delete a site

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**.
5. Select the site and choose **Delete**.
6. In the confirmation dialog box, choose **Delete**.

Deleting a site using the AWS CLI

Use the [delete-site](#) command.

Add a link using AWS Network Manager

Add a link to associated a device with a site.

To add a link

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**.
5. Choose the link for the site **ID** that you want to add a link to, and then choose the **Links** tab.

Note

Choose the link. Do not select the check box.

6. Choose the **Links** tab, and then choose **Create link**.
7. For **Name** and **Description**, enter a name and description for the link.
8. For **Upload speed (Mbps)**, enter the upload speed in Mbps.
9. For **Download speed (Mbps)**, enter the download speed in Mbps.
10. (Optional) For **Provider**, enter the name of the service provider.
11. (Optional) For **Type**, enter the type of link, for example, **broadband**.
12. (Optional) Under **Additional settings**, add one or more **Key** and **Value Tags** to help further identify this link.
13. Choose **Create link**.

Edit a link using AWS Network Manager

Edit the link between two devices in your Cloud WAN global network.

To update a link

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**.
5. Choose the **Links** tab.
6. On the **Links** page, select the check box for the link that you want to update, and then choose **Edit**.
7. Modify any of the link settings as needed, including adding, editing, or removing tags.
8. Choose **Edit link**.

Delete a link using AWS Network Manager

You can delete the link between two devices without deleting the devices.

To delete a link

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.

2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Sites**.
5. Choose the **Links** tab.
6. On the **Links** page, select the check box for the link that you want to delete, and then choose **Delete**.
7. Confirm that you want to delete the link by choosing **Delete** again.

Devices in AWS Global Networks for Transit Gateways

Devices represent a physical or virtual appliance in AWS Global Networks for Transit Gateways. When you add a device to your core network using AWS Network Manager, you can include optional information such as vendor, model and serial number to help you more easily identify the device.

In addition, you'll indicate whether the device is on-premises or in the AWS Cloud. If the device is on-premises you can specify optional information such as physical address. If the device is in the AWS Cloud, you can specify the zone, subnet ID, latitude and longitude, and physical address. Tags are also used to more help you identify this Network Manager resource.

Once added to your global network, a device can then be associated with a site. Before you can associate the device with a site using a link, you must first create the site. For more information on creating sites and linking the site to a device, see [the section called "Sites and links"](#).

Note

A single device can't be associated with multiple sites.

Topics

- [Add a device using AWS Network Manager](#)
- [Delete a device using AWS Network Manager](#)
- [Edit a device using AWS Network Manager](#)
- [View device details using AWS Network Manager](#)

Add a device using AWS Network Manager

Create a device to represent a physical or virtual appliance.

To add a device

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**. Choose **Create device**.
5. For **Name** and **Description**, enter a name and description for the device.
6. For **Model**, enter the device model number.
7. For **Serial number**, enter the serial number for the device.
8. For **Type**, enter the device type.
9. For **Vendor**, enter the name of the vendor, for example, *Cisco*.
10. For **Location type**, specify whether the device is located in a remote location (on-premises network, data center, or other cloud provider) or in AWS.

If you choose **AWS Cloud**, specify the location of the device within AWS. For **Zone**, specify the name of an Availability Zone, Local Zone, Wavelength Zone, or an Outpost. For **Subnet**, specify the Amazon Resource Name (ARN) of a subnet (for example, `arn:aws:ec2:us-east-1:111111111111:subnet/subnet-abcd1234`).

11. For **Address**, enter the physical address of the site, for example, *New York, NY 10004*.
12. For **Latitude**, enter the latitude coordinates for the site, for example, *40.7128*.
13. For **Longitude**, enter the longitude coordinates for the site, for example, *-74.0060*.
14. Choose **Create device**.

Creating and viewing a device using the AWS CLI

Use the following commands:

- To create a device: [create-device](#)
- To view your devices: [get-devices](#)

Delete a device using AWS Network Manager

Delete any device from your global network that's no longer needed or is no longer available.

To delete a site

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the device that you want to delete, and then choose **Delete**.
6. Confirm that you want to delete the site by choosing **Delete** again.

Deletion occurs immediately.

Edit a device using AWS Network Manager

Edit the details of a device, including whether the location type is either on-premises or AWS Cloud.

To update a device

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the check box of the device that you want to update, and then choose **Edit**.
6. Choose **Edit device**.
7. Add or update any of the following device information:
 - **Description**
 - **Model**
 - **Serial number**

- **Type**
- **Vendor**
- **Location type**
- **Latitude**
- **Longitude**
- **Tags**

8. Choose **Edit device**.

View device details using AWS Network Manager

View details about a device. On the device details page you can access tabs:

- **Overview**

This tab provides general information about the device, such as the device State, Vendor, and Model. You can also edit, delete, and associate or disassociate the device with a site,

- **Links**

Associate or disassociate a link with a device.

- **On-premises associations**

Associate or disassociate a device with a customer gateway. You must have at least one gateway set up and one link to create the association.

- **Connect peer associations**

Associate a Connect peer with a device, allowing you to connect with a transit gateway. You must have at least one Connect peer and one link.

- **Connections**

Create a connection between two devices using a link. You can create a connection between two devices in your global network. The connection can be between a physical or virtual appliance and a third-party appliance in a VPC, or between physical appliances in an on-premises network. A connection is created for a specific global network and cannot be shared with other global networks.

- **VPNs**

View the VPNs associated with the device. On this tab you can only view the associations of a transit gateway with a device.

- **Monitoring**

Monitor the device's data in, data out, and Tunnel down count average with CloudWatch metrics. You can modify the CloudWatch time frame as well as add these metrics to your global network dashboard.

Topics

- [Associate or disassociate a device link using AWS Network Manager](#)
- [Associate or disassociate an on-premises link using AWS Network Manager](#)
- [Associate or disassociate a Connect peer using AWS Network Manager](#)
- [View VPNs using AWS Network Manager](#)
- [Monitor devices using AWS Network Manager](#)

Associate or disassociate a device link using AWS Network Manager

Associate a link with a device in your AWS global network. In order to associate a link with a device, you must first create the link. For more information on creating links, see [the section called “Add a link”](#).

You can only associate one link with one device. If a link is already associated with a device, and you want to use that link with another device, you must first disassociate the link the device it's associated with.

To associate a link with a device

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the link for the device **ID** that you want to add a link to, and then choose the **Links** tab.

Note

Choose the link. Do not select the check box.

6. Choose the **Links** tab, and then choose **Associate link**.
7. Choose the link that you want to associate with the device.
8. Choose **Associate link**.

The link is available to use immediately.

If you to use a link with another device, you must first disassociate the link from its original device.

To disassociate a link from a device

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the link for the device **ID** that you want to add a link to, and then choose the **Links** tab.

Note

Choose the link. Do not select the check box.

6. Choose the **Links** tab, and then choose **Associate link**.
7. Choose the check box for the link that you want to disassociate from a device.
8. Choose **Disassociate link**.

Disassociation occurs immediately.

Associate or disassociate an on-premises link using AWS Network Manager

Associate or disassociate an on-premises device link association in your AWS global network.

You can only associate one link with a customer gateway. If a link is already associated with a customer gateway, and you want to use that link with another gateway, you must first disassociate the link the gateway it's currently associated with.

To create an on-premises association

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the link for the device **ID** that you want to create an on-premises association for.
6. Choose the **On-premises associations** tab.
7. Choose **Associate**.
8. Choose the on-premises **Customer gateway**.
9. (Optional) Choose the **Link** used for the connection.
10. Choose **Create on-premises association**.

The link is available to use immediately.

To disassociate an on-premises association

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the device **ID** link.
6. Choose the **On-premises association** tab.
7. Choose the check box for the on-premises association that you want to disassociate.
8. Choose **Disassociate**.

Disassociation occurs immediately.

Associate or disassociate a Connect peer using AWS Network Manager

Associate or disassociate a Connect peer device link association in your AWS global network.

You can only associate one link with a Connect peer. If a link is already associated with a Connect peer, and you want to use that link with another Connect peer, you must first disassociate the link the Connect peer it's associated with.

To create a Connect peer association

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the link for the device **ID** that you want to create an on-premises association for.
6. Choose the **Connect peer** tab.
7. Choose **Associate**.
8. Choose the on-premises **Connect peer**.
9. (Optional) Choose the **Link** used for the connection.
10. Choose **Create Connect peer association**.

The link is available to use immediately.

To disassociate a Connect peer association

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the device **ID** link.
6. Choose the **Connect peer** tab.
7. Choose the check box for the Connect peer that you want to disassociate.

8. Choose **Disassociate**.

Disassociation occurs immediately.

View VPNs using AWS Network Manager

The VPNs page displays a list of your VPN connections for a device.

To view device VPN connections

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the device that you want to view the VPN connections for.
6. Choose **VPNs**.

Monitor devices using AWS Network Manager

Monitor device Amazon CloudWatch events on the Network Manager Monitoring page.

To monitor devices

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**.
5. Choose the **Monitoring** tab.
6. The **Monitoring** page displays data for the following:
 - **Data In**
 - **Data Out**
 - **Tunnel down count Average**

(Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Connections in AWS Global Networks for Transit Gateways

Create a connection between two devices in your global network using AWS Network Manager. The connection can be between a physical or virtual appliance and a third-party appliance in a VPC, or between physical appliances in an on-premises network. To create a connection the device must first be added to your global network. For the steps to add a device, see [the section called “Add a device”](#). You can also use an optional link to create the connection. For the steps to create a link, see [the section called “Add a link”](#).

A connection is created for a specific global network and cannot be shared with other global networks.

Tasks

- [Create a connection using AWS Network Manager](#)
- [Update a connection using AWS Network Manager](#)
- [Delete a connection using AWS Network Manager](#)

Create a connection using AWS Network Manager

Create a connection between two existing devices in your global network.

To create a connection

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and choose the ID of the device.
5. Choose **Connections**, and then choose **Create connection**.
6. For **Name** and **Description**, enter a name and description for the connection.
7. (Optional) For **Link**, choose a link to associate with the first device in the connection.
8. For **Connected device**, choose the ID of the second device in the connection.
9. (Optional) For **Connected link**, choose a link to associate with the second device in the connection.
10. Choose **Create connection**.

To create a connection using the AWS CLI

Use the [create-connection](#) command.

Update a connection using AWS Network Manager

You can update the information for an existing connection.

To update a connection

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and select the device.
5. Choose **Connections**, and select the connection.
6. Choose **Edit**.
7. Update the connection details as needed, and then choose **Edit connection**.

To update a connection using the AWS CLI

Use the [update-connection](#) command.

Delete a connection using AWS Network Manager

If you no longer need a connection, you can delete it.

To delete a connection

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and select the device.
5. Choose **Connections**, and select the connection.
6. Choose **Delete**.
7. When prompted for confirmation, choose **Delete**.

To delete a connection using the AWS CLI

Use the [delete-connection](#) command.

Gateway associations in AWS Global Networks for Transit Gateways

Create a customer gateway association with either a device or with a transit gateway Connect peer.

Customer gateway associations

To add your on-premises network to your global network, you associate a customer gateway with your device, and optionally, a link. The customer gateway must already be in your global network as part of a VPN attachment in your transit gateway. If you specify a link, it must already be associated with the specified device.

For more information about creating a customer gateway, see [Create a Customer Gateway](#) in the *AWS Site-to-Site VPN User Guide*. For more information about creating a VPN attachment to a transit gateway, see [Transit Gateway VPN Attachments](#) in *Amazon VPC Transit Gateways*.

For more information about viewing the topology of your on-premises network in Network Manager, see [the section called “Access transit gateway network dashboards”](#) />.

Transit Gateway Connect peer associations

You can associate a [Connect peer](#) (in a transit gateway Connect attachment) with a device, and optionally, with a link.

If you specify a link, it must be associated with the specified device.

Topics

- [Associate a customer gateway using AWS Network Manager](#)
- [Disassociate a customer gateway using AWS Network Manager](#)
- [Add a Connect peer association using AWS Network Manager](#)
- [Disassociate a Connect peer using AWS Network Manager](#)

Associate a customer gateway using AWS Network Manager

You can associate a customer gateway with a device and link using the Network Manager console on either of the following pages:

- On the **Transit gateways** page
- On the **Devices** page

Transit gateways page

To associate a customer gateway using the Transit gateways page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose the ID of your transit gateway.
5. Choose **On-premises associations**.

6. Select your customer gateway and choose **Associate**.
7. For **Device**, select the ID of the device to associate. For **Link**, select the ID of the link to associate.
8. Choose **Edit on-premises association**.

Devices page

To associate a customer gateway using the Devices page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and then choose the ID of your device.
5. Choose **On-premises associations**.
6. Choose **Associate**.
7. For **Customer gateway**, select the ID of the customer gateway to associate. For **Link**, select the ID of the link to associate.
8. Choose **Create on-premises association**.

Create a customer gateway association using the AWS CLI

You can view and create a customer gateway association using the following commands.

- To associate a customer gateway with a device and link: [associate-customer-gateway](#)
- To view your customer gateway associations: [get-customer-gateway-associations](#)

Disassociate a customer gateway using AWS Network Manager

You can disassociate a customer gateway from a device or link using the Network Manager console on either of the following pages:

- On the **Transit gateways** page
- On the **Devices** page

Transit gateways page

To disassociate a customer gateway using the Transit gateways page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose **On-premises associations**.
5. Select your customer gateway and choose **Disassociate**.

Devices page

To disassociate a customer gateway using the Devices page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and then choose the ID of your device.
5. Choose **On-premises associations**.
6. Select your customer gateway and choose **Disassociate**.

Disassociate a customer gateway association using the AWS CLI

You can view and disassociate a customer gateway association using the following command.

- To view your customer gateway associations: [get-customer-gateway-associations](#)
- To disassociate a customer gateway from a device and link: [disassociate-customer-gateway](#)

Add a Connect peer association using AWS Network Manager

Create a transit gateway Connect peer association using the Network Manager console on either of the following pages:

- On the **Transit gateways** page
- On the **Devices** page

Transit gateways page

To associate a Connect peer using the Transit gateways page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose the ID of your transit gateway.
5. Choose **Connect peer associations**.
6. Select the Connect peer and choose **Edit**.
7. For **Device**, select the ID of the device to associate. For **Link**, select the ID of the link to associate.
8. Choose **Edit Connect peer association**.

Devices page

To associate a Connect peer using the Devices page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and choose the ID of the device.
5. Choose **Connect peer associations**.
6. Choose **Associate**.
7. For **Connect peer**, choose the Connect peer.
8. (Optional) For **Link**, choose the link for the Connect peer association.
9. Choose **Create Connect peer association**.

Working with Connect peer associations using the AWS CLI

You can view and create Connect peer associations using the following commands.

- To associate a Connect peer with a device: [associate-transit-gateway-connect-peer](#)
- To view your Connect peer associations: [get-transit-gateway-connect-peer-associations](#)

Disassociate a Connect peer using AWS Network Manager

You can disassociate a Connect peer from a device in one of the following ways:

- On the **Transit gateways** page
- On the **Devices** page

Transit gateways page

To disassociate a Connect peer using the Transit gateways page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**, and then choose **Connect peer associations**.
5. Select the Connect peer and choose **Disassociate**.

Devices page

To disassociate a Connect peer using the Devices page

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Devices**, and then choose the ID of your device.
5. Choose **Connect peer associations**.

6. Select the Connect peer and choose **Disassociate**.

Working with Connect peer associations using the AWS CLI

You can view and disassociate Connect peer associations using the following commands.

- To view your Connect peer associations: [get-transit-gateway-connect-peer-associations](#)
- To disassociate a Connect peer from a device: [disassociate-transit-gateway-connect-peer](#)

Resource tags in AWS Global Networks for Transit Gateways

A *tag* is a metadata label that either you or AWS assigns to an AWS resource. Each tag consists of a *key* and a *value*. For tags that you assign, you define the key and the value. For example, you might define the key as `purpose` and the value as `test` for one resource.

Tags help you do the following:

- Identify and organize your AWS resources. Many AWS services support tagging, so you can assign the same tag to resources from different services to indicate that the resources are related.
- Control access to your AWS resources. For more information, see [Controlling access to AWS resources using tags](#) in the *IAM User Guide*.

Supported resources

The following global networks resources support tagging:

- Global networks
- Devices
- Sites
- Links

Tagging restrictions

The following basic restrictions apply to tags on global networks resources:

- Maximum number of tags that you can assign to a resource: 200

- Maximum key length: 128 Unicode characters
- Maximum value length: 256 Unicode characters
- Valid characters for key and value: a-z, A-Z, 0-9, space, and the following characters: _ . : / = + - and @
- Keys and values are case sensitive
- You cannot use `aws :` as a prefix for keys; it's reserved for AWS use

AWS Global Networks for Transit Gateways dashboards

The AWS Network Manager console uses dashboard visualizations to help you view and monitor all aspects of your transit gateway networks and transit gateways. Some of the dashboards include:

- World maps that pinpoint where your network resources, such as edge locations, devices, and attachments, are located.
- Monitoring that uses CloudWatch to track 15-months' worth of statistics, giving you a better perspective on how your networks are performing.
- Event tracking that streams real-time events to an events dashboard.
- Topological and logical diagrams of your transit gateway networks and transit gateways.

There are separate dashboards for your transit gateway networks and transit gateways.

Topics

- [Access transit gateway network dashboards using AWS Network Manager](#)
- [Access transit gateway dashboards using AWS Network Manager](#)

Access transit gateway network dashboards using AWS Network Manager

The AWS Network Manager console provides a group of dashboards for AWS Global Networks for Transit Gateways, allowing you to view and monitor your network of transit gateways. Dashboards include information about network resources, their geographic locations, the network topology, and the logical network associations. If you want to view the dashboards for a specific transit gateway, see [the section called "Access transit gateway dashboards"](#).

Topics

- [Overview](#)
- [Geography](#)
- [Topology tree](#)
- [Events](#)
- [Monitoring](#)

- [Route analyzer](#)

Overview

The Overview page displays details about your transit gateway network, the VPN status, the Connect peer status, and any network events affecting your transit gateways.

To access transit gateway network details

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. On the **Overview** page you contains the following information:
 - Your transit gateway network **Inventory**:

Description

Transit gateways

The total number of registered transit gateways in your global network. Choose the link to open the **Transit gateways** page to view more information about your transit gateways.

Sites

The total number of sites associated with your transit gateways. Choose the link to open the **Sites** page to view more information about your transit gateway sites.

Devices

The total number of devices associated with your transit gateways. Choose the link to open the **Devices** page to view more information about your transit gateway devices.

- The **Transit gateways VPN status**. The following is displayed:
 - **ID** – The ID of the transit gateway. Choose the link to open details about the transit gateway.
 - **Name** – Name of the transit gateway.
 - **Region** – Region where the transit gateway is located
 - **Down VPN** – The percentage of your total transit gateway VPNs that are down.
 - **Impaired VPN** – The percentage of your total VPNs that are impaired.
 - **Up VPN** – The percentage of your total VPNs that are up.
- The **Transit gateways connect peer status**. The following is displayed:
 - **ID** – The ID of the transit gateway.
 - **Name** – Name of the transit gateway.
 - **Region** – Region where the transit peer is located
 - **Down Connect peer** – The percentage of your total transit gateway Connect peers that are down.
 - **Impaired Connect peer** – The percentage of your total transit gateway Connect peers that are impaired.
 - **Up VPN** – The percentage of your total transit gateway Connect peers that are up.
- The **Network events summary** displays CloudWatch Events number of core network attachments per edge, shown as a stacked column chart.

(Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Geography

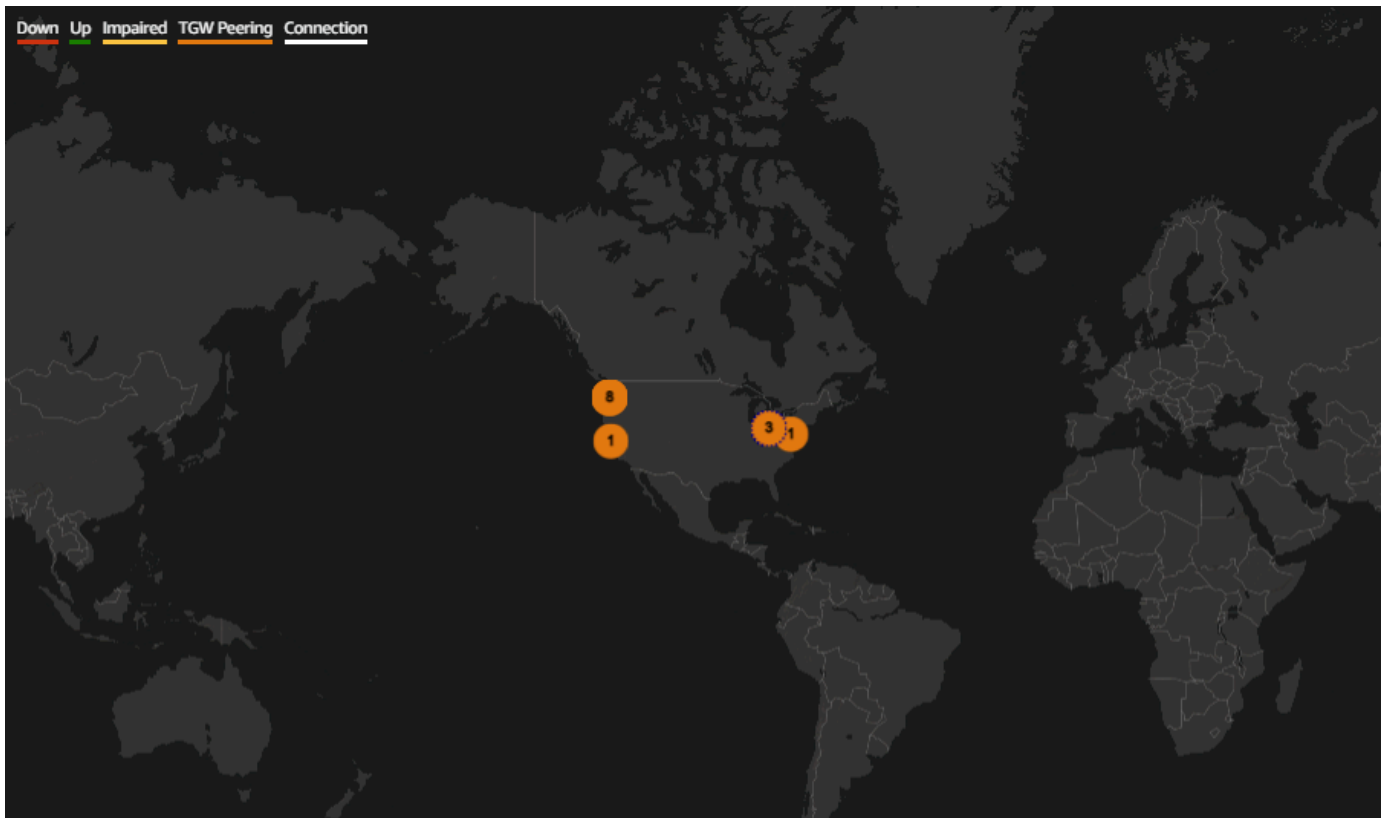
The Geography page displays a world map showing the locations of your transit gateway network.

To access a geographic map of your transit gateways

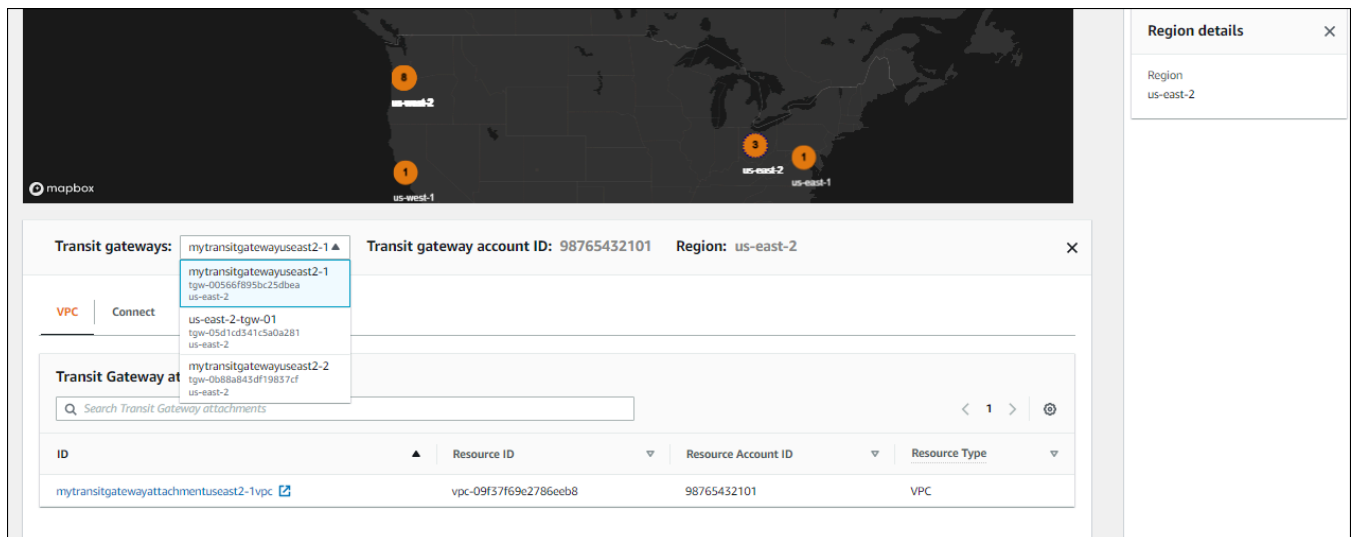
1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Geography** tab.

A world map displays, showing you the locations of the following:

- **AWS TGWs and VPCs.**
 - The **Connectivity** of **VPNs**, **Direct Connects**, and **Connect peers**.
 - **On-premises Sites** and **Devices**.
 - **Not associated Sites** and **Devices**.
7. In the following example, there are four AWS Regions, **us-west-1**, **us-west-2**, **us-east-1**, and **us-east-2**. Each Region is labeled and represented by a number, indicating the number of transit gateways in that Region. For example, **us-east-2** is represented by the number 3, indicating that there are three network resources associated with the us-west-2 Region.



8. If your account is a delegated administrator in a multi-account environment, you can view details about the transit gateways for different accounts.
9. Choose the number representing a Region. For example, choose 3. The following information displays:
 - The right pane shows the AWS Region, us-east-2.
 - A bottom panel shows with a **Transit Gateways** dropdown list option, displaying each transit gateway in that Region. In this example, there are 3 transit gateways in us-east-2. Choose a transit gateway from the dropdown list to view details about that transit gateway. In this example, you can see that the **Resource Account ID** for this transit gateway is another account in the multi-account environment, 98765432101.



- To view more details about the transit gateway, choose the ID link to open the **Transit gateway details** page for the gateway.

If your global network is part of a multi-account environment, you can choose an **ID** from a member account and view details about that attachment. The **Resource Account ID** column displays the account ID that the transit gateway belongs to.

Viewing details about a member's resources prompts you to use the Network Manager console to switch roles to the member account where the resource is located.

Note

Switching roles logs you out of the current account and into the member account associated with the attachment.

Switch global networks console roles to view resource details

To view resource details in a member account

- When choosing a link to a member account, you're prompted to switch console roles:

Switch Role

Allows management of resources across Amazon Web Services accounts using a single user ID and password. You can switch roles after an Amazon Web Services administrator has configured a role and given you the account and role details. [Learn more.](#)

Account* ⓘ

Role* ⓘ

Display Name ⓘ

Color a a a a a a

***Required** [Cancel](#) [Switch Role](#)

2. The following values populate the **Switch Role** screen. Keep the following values:

- **Account** — The account ID for the member account that the resource is associated with.
- **Role** — `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess` is the required IAM role for accessing resources across multiple accounts.

3. Choose **Switch Role**.

You're logged out of your current account and into that member account. A new tab opens showing the details of the resource. For example, if you choose a VPC resource, the VPC resource page opens for the member account that owns the resource.

4. Depending on the delegated permission level assigned to the delegated administrators and the management account when trusted access was enabled, you can either view information (read-only permission) about the resource or add/modify (administrator permission) the resource.
5. To return to the original member account, choose one of the following:
- On your current tab, choose the browser **Back** button. On the **Switch Role** login screen, enter the **Account** ID of the account you want, and then choose **Switch Role**.

- If you haven't closed it, choose the tab for the account you've just logged out of, and then choose **Reload**.

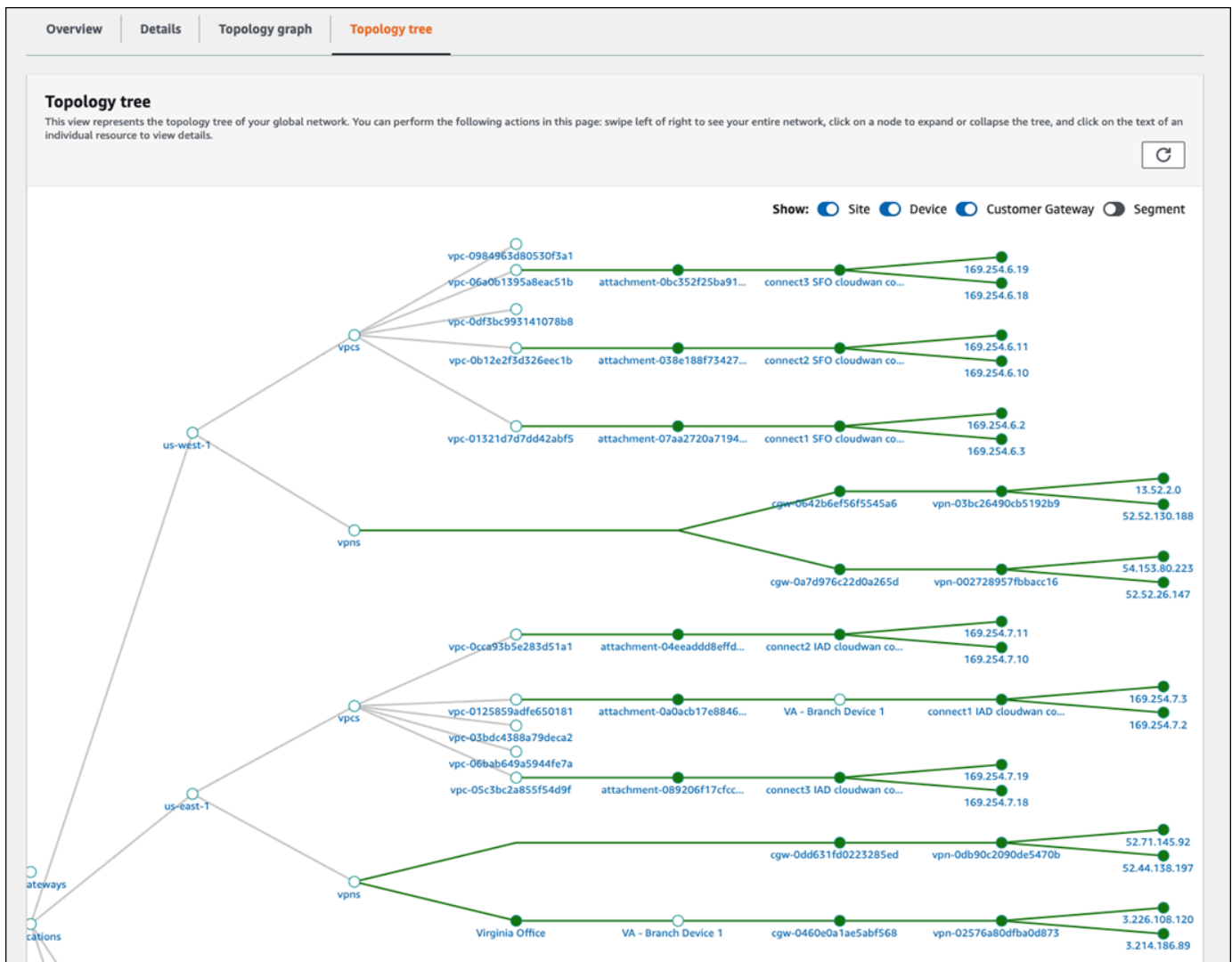
Topology tree

The **Topology tree** page shows a logical diagram of your transit gateway network.

To access the topology tree for a transit gateway network

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Topology tree** tab.
7. By default, the **Topology tree** page displays all **Sites**, **Devices**, and **Customer Gateways** of your transit gateway and the logical relationships between them. You can filter the network tree to show specific resources types only to view information about the specific resource it represents. The line colors represent the state of the relationships between AWS and the on-premises resources.

The following example shows the topology tree for two edge locations, **us-west-1** and **us-east-1**.



8. In the **Topology tree**, choose an attachment. The attachment details display in the left pane.
9. If your global network is part of a multi-account environment, you can choose a **Resource ID** from a member account and view details about that attachment.

Viewing details about a member's resources prompts you to switch Network Manager console roles to the member account where the resource is located.

Note

Switching roles logs you out of the current account Device and into the delegated administrator account associated with the attachment.

Switch global networks console roles to view resource details

To view resource details in a member account

1. When choosing a link to a member account, you're prompted to switch console roles:

Switch Role

Allows management of resources across Amazon Web Services accounts using a single user ID and password. You can switch roles after an Amazon Web Services administrator has configured a role and given you the account and role details. [Learn more.](#)

Account* ⓘ

Role* ⓘ

Display Name ⓘ

Color a a a a a a

***Required** Cancel Switch Role

2. The following values populate the **Switch Role** screen. Keep the following values:
 - **Account** — The account ID for the member account that the resource is associated with.
 - **Role** — IAMRoleForAWSNetworkManagerCrossAccountResourceAccess is the required IAM role for accessing resources across multiple accounts.

3. Choose **Switch Role**.

You're logged out of your current account and into that member account. A new tab opens showing the details of the resource. For example, if you choose a VPC resource, the VPC resource page opens for the member account that owns the resource.

4. Depending on the delegated permission level assigned to the delegated administrators and the management account when trusted access was enabled, you can either view information (read-only permission) about the resource or add/modify (administrator permission) the resource.
5. To return to the original member account, choose one of the following:

- On your current tab, choose the browser **Back** button. On the **Switch Role** login screen, enter the **Account** ID of the account you want, and then choose **Switch Role**.
- If you haven't closed it, choose the tab for the account you've just logged out of, and then choose **Reload**.

Events

Track your transit gateway events using Amazon EventBridge that delivers a near-real-time stream of system events that describe changes in your resources. Using simple rules that you can quickly set up, you can match events and route them to one or more target functions or streams. For more information, see the [Amazon EventBridge User Guide](#).

To access transit gateway network events

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Events** tab.

The **Events** section updates with the events that occurred during the time frame.

(Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Monitoring

You can monitor your transit gateways using Amazon CloudWatch which collects raw data and processes it into readable, near-real-time metrics. These statistics are kept for 15 months, so that you can access historical information and gain a better perspective on how your network is performing. You can also set alarms that watch for certain thresholds, and send notifications or take actions when those thresholds are met. For more information, see the [Amazon CloudWatch User Guide](#).

On the monitoring page you can view usage metrics for your transit gateways, filtering by specific transit gateways.

To access transit gateway network monitoring details

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Monitoring** tab.
7. Choose a transit gateway that you want to monitor.

If you're using an account that's set up as a delegated administrator between accounts, you can choose a transit gateway from one of those other accounts. The transit gateway list displays the ID, the Region, and the account ID.

8. (Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

9. The page updates the following transit gateway monitors:

- Bytes in
- Bytes out
- Bytes dropped – black hole
- Bytes dropped – no route
- Packets in
- Packets out
- Packets dropped – black hole
- Packets dropped – no route

10. (Optional) Choose **Add to dashboard** to add this metric to your CloudWatch dashboard.

For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the Amazon CloudWatch User Guide.

Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Route analyzer

The Route Analyzer analyzes the routing path between a specified source and destination.

Note

Route Analyzer checks the routes on Transit Gateway route tables only

To analyze transit gateway routes

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Route Analyzer** tab.
7. In the **Source** section,

- Choose the source **Transit Gateway** for the route that you want to analyze.

If you're logged on to an account that's set up as a delegated administrator between accounts, you can choose a transit gateway from one of those other accounts. The transit gateway list displays the ID, the Region, and the account ID.

- Choose the source **Transit Gateway attachment** for the route.
 - Enter either the IPv4 or IPv6 **IP address**.
 - Clear the **Include return path in results** check box if you don't want . This is chosen by default.
 - Choose if this is a **Middlebox appliance**. For more information on middlebox configurations, see [Route analysis with a middlebox configuration](#).
8. In the Destination section,
 - Choose the destination **Transit Gateway**.
 9. Choose **Run route analysis**.

If you're logged on to an account that's set up as a delegated administrator between accounts, you can choose a transit gateway from one of those other accounts. The transit gateway list displays the ID, the Region, and the account ID.

- Choose the destination **Transit Gateway attachment** for the route.
 - Enter either the IPv4 or IPv6 **IP address**.
10. The Results of route analysis return the **Source** and **Destination** transit gateways and the current **Status**. An error message is returned if no information is found in the transit gateway route table. For more information on route tables, see [Transit gateway route tables](#).

Access transit gateway dashboards using AWS Network Manager

The AWS Network Manager console provides a group of dashboards for AWS Global Networks for Transit Gateways, allowing you to view and monitor your transit gateways. Dashboards include information about network resources, their geographic locations, the network topology, and the logical network associations. If you want to view the dashboards for all transit gateways in your global network, see [the section called “Access transit gateway network dashboards”](#).

Topics

- [Overview](#)
- [Topology tree](#)
- [Events](#)
- [Monitoring](#)
- [On-premises associations](#)
- [Connect peer associations](#)

Overview

To access the transit gateway resource inventory

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity** choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway networks**.
5. The **Transit gateways** page opens, showing a list of your transit gateways.
6. Choose the **ID** of the transit gateway you want to see more information about.
7. On the **Overview** page you can view the following information:
 - Your transit gateway details.
 - The transit gateway attachments, along with information about each of those attachments.

Use the following legend to understand the icons on this page:

Description

VPC

The total number of VPC attachments in your transit gateway network.

VPN

The total number of VPN attachments in your transit gateway.

Direct Connect Gateway

The total number of Direct Connect gateways attached to your transit gateway.

Connect

The total number of Connect peer attachments in your transit gateway.

Transit Gateway

The total number of Transit Gateways.

8. The **Details** section shows information about your global network: the transit gateway **ID**, its **Name**, the **Region** where it's located, and the current **State** of the gateway.

Note

To see details about a different transit gateway, choose the dropdown list and then choose the transit gateway.

9. The **Transit Gateway attachment** section displays details about your attachments: the Transit Gateway **ID**, the **Resource ID**, and the **Resource Type**.
10. The **VPNs** section displays details about your VPN attachments: the VPN **ID**, the **Device** using the VPN attachment, and any **Link** associated with the attachment.
11. The **Connect peers** section displays details about your Connect peer attachments: the name of the **Connect peer** and the **Device** using that Connect peer.
12. The **Network events summary** section shows the network events for that transit gateway. You must first onboard to see network events. Choose **Onboard CloudWatch Insights** to enable viewing network events.

13. (Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Topology tree

The **Topology tree** page shows a logical diagram of your transit gateways.

To view a transit gateway topology tree

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity** choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway networks**.
5. The **Transit gateways** page opens, showing a list of your transit gateways.
6. Choose the **ID** of the transit gateway you want to see more information about.
7. Choose the **Topology tree** tab.
8. By default, the **Topology tree** page displays all **Sites**, **Devices**, and **Customer Gateways** of your transit gateway and the logical relationships between them. You can filter the network tree to show specific resource types only to view information about the specific resource it represents. The line colors represent the state of the relationships between AWS and the on-premises resources.
9. In the **Topology tree**, choose a resource. The resource details display in the right pane.
10. If your global network is part of a multi-account environment, you can choose a **Resource ID** from a member account and view details about that attachment.

Viewing details about a member's resources prompts you to switch Network Manager console roles to the member account where the resource is located.

Note

Switching roles logs you out of the current account and into the delegated administrator account associated with the attachment.

Switch global networks console roles to view resource details

To view resource details in a member account

1. When choosing a link to a member account, you're prompted to switch console roles:

Switch Role

Allows management of resources across Amazon Web Services accounts using a single user ID and password. You can switch roles after an Amazon Web Services administrator has configured a role and given you the account and role details. [Learn more.](#)

Account* ⓘ

Role* ⓘ

Display Name ⓘ

Color

***Required** Cancel Switch Role

2. The following values populate the **Switch Role** screen. Keep the following values:
 - **Account** — The account ID for the member account that the resource is associated with.
 - **Role** — `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess` is the required IAM role for accessing resources across multiple accounts.

3. Choose **Switch Role**.

You're logged out of your current account and into that member account. A new tab opens showing the details of the resource. For example, if you choose a VPC resource, the VPC resource page opens for the member account that owns the resource.

4. Depending on the delegated permission level assigned to the delegated administrators and the management account when trusted access was enabled, you can either view information (read-only permission) about the resource or add/modify (administrator permission) the resource.
5. To return to the original member account, choose one of the following:
 - On your current tab, choose the browser **Back** button. On the **Switch Role** login screen, enter the **Account** ID of the account you want, and then choose **Switch Role**.
 - If you haven't closed it, choose the tab for the account you've just logged out of, and then choose **Reload**.

Events

Track your transit gateway events using Amazon EventBridge that delivers a near-real-time stream of system events that describe changes in your resources. Using simple rules that you can quickly set up, you can match events and route them to one or more target functions or streams. For more information, see the [Amazon EventBridge User Guide](#).

To track transit gateway events

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Events** tab.

The **Events** section updates with the events that occurred during the time frame.

(Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Monitoring

You can monitor your transit gateways using Amazon CloudWatch, which collects raw data and processes it into readable, near-real-time metrics. These statistics are kept for 15 months, so that you can access historical information and gain a better perspective on how your network is performing. You can also set alarms that watch for certain thresholds, and send notifications or take actions when those thresholds are met. For more information, see the [Amazon CloudWatch User Guide](#).

On the monitoring page you can view usage metrics for your transit gateways, filtering by specific transit gateways.

To view transit monitoring details

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity** choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway networks**.
5. The **Transit gateways** page opens, showing a list of your transit gateways.
6. Choose the **ID** of the transit gateway you want to see more information about.
7. Choose the **Monitoring** tab.

8. If you want to choose a different transit gateway to monitor, choose that transit gateway from the list.
9. (Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

10. The page updates the following transit gateway monitors:
 - **Bytes in**
 - **Bytes out**
 - **Bytes dropped – black hole**
 - **Bytes dropped – no route**
 - **Packets in**
 - **Packets out**
 - **Packets dropped – black hole**
 - **Packets dropped – no route**
11. (Optional) Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

 Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

On-premises associations

The **On-premises** page displays information about your on-premises devices for this transit gateway. On this page you can associate or disassociate any of your devices..

To view on-premises associations

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity** choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway networks**.
5. The **Transit gateways** page opens, showing a list of your transit gateways.
6. Choose the **ID** of the transit gateway you want to see more information about.
7. Choose the **On-premises associations** tab.
8. The **Transit Gateway** on-premises association page displays the **Customer gateway**, **Device**, **Link**, and **State** of the transit gateway.

To associate a device

1. Choose the **Customer gateway** you want to associate a device with.
2. Choose **Associate**.
3. On the **Edit on-premises association** page, choose the **Device** and optional **Link** for the association.
4. Choose **Edit on-premises association**.

To disassociate an on-premises device

1. Choose the **Customer gateway** you want to disassociate.
2. Choose **Disassociate**.

Connect peer associations

The Connect peer associations page displays information about your Connect peers for this transit gateway. You can also disassociate any of your devices.

To access Connect peer associations

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity** choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway networks**.
5. The **Transit gateways** page opens, showing a list of your transit gateways.
6. Choose the **ID** of the transit gateway you want to see more information about.
7. Choose the **Connect peer associations** tab.
8. The **Connect peer associations** page displays the **Connect peer**, **Device**, **Link**, and **State** of the transit gateway.

To disassociate a Connect peer device

1. Choose the **Connect peer** you want to disassociate.
2. Choose **Disassociate**.

Route Analyzer for AWS Network Manager

With AWS Global Networks for Transit Gateways, you can use the Route Analyzer to perform an analysis of the routes in your transit gateway route tables. Through AWS Network Manager, Route Analyzer analyzes the routing path between a specified source and destination, and returns information about the connectivity between components. You can use the Route Analyzer to do the following:

- Verify that the transit gateway route table configuration will work as expected before you start sending traffic.
- Validate your existing route configuration.
- Diagnose route-related issues that are causing traffic disruption in your global network.

Note

Route Analyzer does not work with intra-Region peering.

Topics

- [Route Analyzer basics](#)
- [Perform a route analysis](#)
- [Example: Route analysis for peered transit gateways](#)
- [Example: Route analysis with a middlebox configuration](#)

Route Analyzer basics

To use the Route Analyzer, you indicate the path for the traffic from a source to a destination. For the source, you specify the transit gateway, the transit gateway attachment from which the traffic originates, and a source IPv4 or IPv6 address. The Route Analyzer analyzes the routes in the associated transit gateway route table for the transit gateway attachment. For the destination, you specify a target IPv4 or IPv6 address, and the destination transit gateway and transit gateway attachment.

If you've configured a middlebox appliance in your VPC, you can indicate the location of the appliance in the route analysis. This enables you to specify multiple network hops in a route

between a source and destination, to help you analyze the route of the traffic. We store this information for use in future analyses. You can update your middlebox appliances later on as needed.

You can also analyze the return path for traffic from the specified destination back to the source.

The following rules apply when using the Route Analyzer:

- The Route Analyzer analyzes routes in transit gateway route tables only. It does not analyze routes in VPC route tables or in your customer gateway devices.
- The transit gateways must be registered in your global network.
- The Route Analyzer does not analyze security group rules or network ACL rules. To capture information about accepted and rejected IP traffic in your VPC, you can use [VPC flow logs](#).
- The Route Analyzer only returns information for the return path if it can successfully return information for the forward path.

Perform a route analysis

Perform a route analysis of your AWS global network. You can only use Route Analyzer using the AWS Global Networks for Transit Gateways console.

To analyze your routes

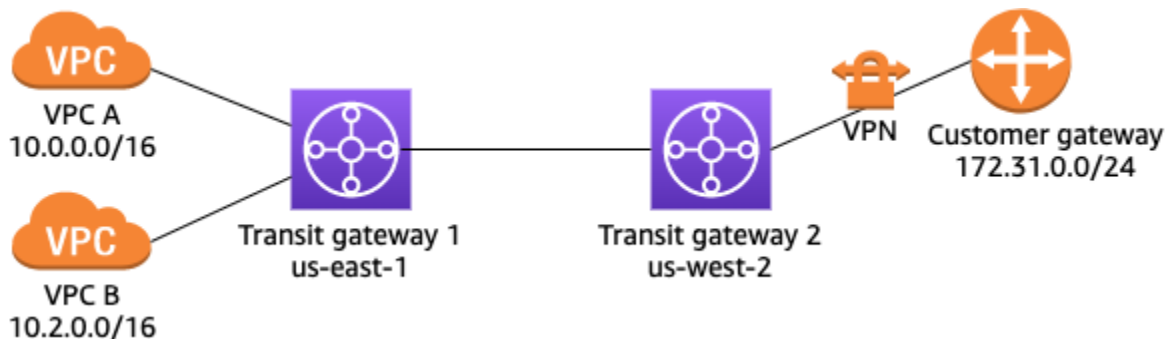
1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit Gateway network**.
5. The **Overview** page opens by default, showing information about your transit gateways.
6. Choose the **Route Analyzer** tab.
7. Under **Source**, do the following:
 - Choose the transit gateway and the transit gateway attachment.
 - For **IP address**, enter a source IPv4 or IPv6 address.
8. Under **Destination**, do the following:
 - Choose the transit gateway and the transit gateway attachment.

- For **IP address**, enter a target IPv4 or IPv6 address.
9. (Optional) To analyze the return path, ensure that you enable **Include return path in results**. If enabled, you must specify an IP address under **Source**.
 10. To specify middlebox appliances in the routing path, choose **Middlebox appliance?**. We store this information for use in future analyses. You can update your middlebox appliances later on as needed.
 11. Choose **Run route analysis**.
 12. The results are displayed under **Results of route analysis**. If you specified **Middlebox appliance?**, choose **Yes** or **No** for each of the attachments to indicate the location of the appliances and to complete the route analysis.

You can choose the ID of any of the resources in the path to view more information about the resources.

Example: Route analysis for peered transit gateways

In the following example, transit gateway 1 has two VPC attachments, and a peering attachment to transit gateway 2. Transit gateway 2 has a Site-to-Site VPN attachment to your on-premises network. You want to use the Route Analyzer to ensure that the VPCs and Site-to-Site VPN connections can route traffic to each other through the transit gateways.



In the Route Analyzer, do the following:

1. Under **Source**, specify transit gateway 1 and the transit gateway attachment for VPC A. Specify an IP address from the CIDR block of VPC A, for example, 10.0.0.7.
2. Under **Destination**, specify transit gateway 2 and the VPN attachment. Specify an IP address from the range of the on-premises network, for example, 172.31.0.8.
3. Ensure that **Include return path in results** is selected.

- Run the route analysis. In the results, verify the path between the source and destination. For example, the following results indicate that there is a forward path from transit gateway 1 to transit gateway 2, but no return path. Check the route table for transit gateway 2, and ensure that there is a static route that points to the peering attachment.

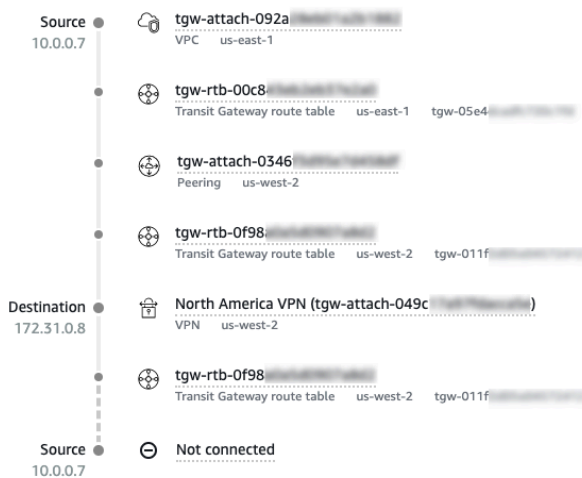
Forward path

Source	Destination	Status
tgw-attach-092a	tgw-attach-049c	Connected

Return path

Source	Destination	Status
tgw-attach-049c	tgw-attach-092a	Not connected

There is no matching route for destination in Transit Gateway route table [tgw-rtb-0f98](#) of Transit Gateway [tgw-011f](#) in region us-west-2.

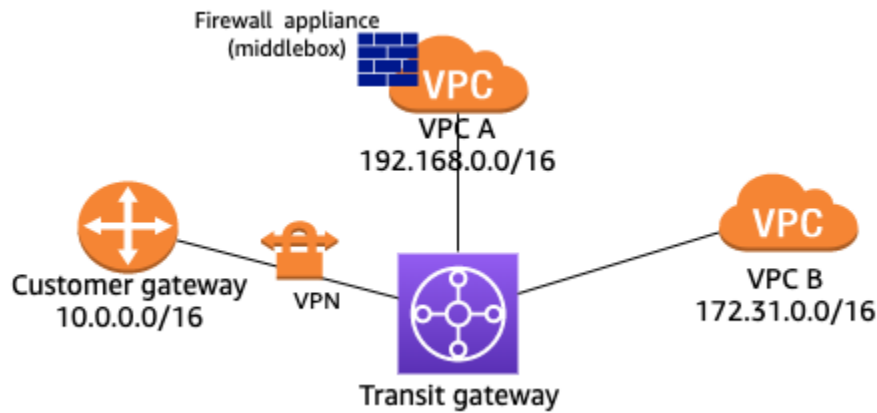


- To run the analysis between VPC B and the VPN connection, modify the information under **Source**. Choose the transit gateway attachment for VPC B, and specify an IP address from the CIDR block of VPC B, for example, `10.2.0.9`.
- Reload the results and verify the path between the source and destination.

For more information about the routing configuration for this scenario, see the [transit gateway peering example](#).

Example: Route analysis with a middlebox configuration

If you've configured a VPC to act as a middlebox appliance for inspecting traffic that flows to other parts of your network, you can indicate the location of the appliance in the route analysis. In the following example, the transit gateway has two VPC attachments and a VPN attachment. VPC A runs a firewall appliance (middlebox) that inspects the traffic that flows between the VPN connection and VPC B.



In the Route Analyzer, you can specify the location of the middlebox appliance as follows:

1. Under **Source**, specify the transit gateway and the VPN attachment. Specify an IP address from the range of the on-premises network, for example, `10.0.0.7`.
2. Under **Destination**, specify the transit gateway and the attachment for VPC B. Specify an IP address from the CIDR block of VPC B, for example, `172.31.0.8`.
3. For **Middlebox appliance?**, choose **Include**.
4. Run the route analysis.
5. For the **Middlebox appliance?** sections for the transit gateway attachment for VPC A, choose **Yes**.

You can choose the ID of any resource in the path to view more information about that resource.

Forward path

Source	Destination	Status
tgw-attach-Oba6	tgw-attach-0c4f	Connected

Return path

Source	Destination	Status
tgw-attach-0c4f	tgw-attach-Oba6	Pending actions

Source
10.0.0.7

tgw-attach-Oba6
VPN us-east-1

tgw-rtb-02dc
Transit Gateway route table us-east-1 tgw-0cb1

VPC A attachment (tgw-attach-08a2)
VPC us-east-1
Middlebox appliance? Yes [Change](#)

tgw-rtb-0669
Transit Gateway route table us-east-1 tgw-0cb1

Destination
172.31.0.8

VPC B attachment (tgw-attach-0c4f)
VPC us-east-1

tgw-rtb-047a
Transit Gateway route table us-east-1 tgw-0cb1

VPC A attachment (tgw-attach-08a2)
VPC us-east-1

Middlebox appliance? [Select](#)

More information

Resource ID
vpc-0780

Transit Gateway attachment
VPC A attachment (tgw-attach-08a2)

Region
us-east-1

Resource Type
VPC

Amazon CloudWatch metrics and events

AWS provides the following monitoring tools to watch the resources in your global network, report when something is wrong, and take automatic actions when appropriate.

- *Amazon CloudWatch* monitors your AWS resources and the applications that you run on AWS in real time. You can collect and track metrics, create customized dashboards, and set alarms that notify you or take actions when a specified metric reaches a threshold that you specify. For more information, see the [Amazon CloudWatch User Guide](#).
- *Amazon EventBridge* delivers a near-real-time stream of system events that describe changes in AWS resources. EventBridge enables automated event-driven computing, as you can write rules that watch for certain events and trigger automated actions in other AWS services when these events happen. For more information, see the [Amazon EventBridge User Guide](#).
- *AWS CloudTrail* provides a record of actions taken by a user, role, or an AWS services in your global network, capturing all API calls for global network events.

Topics

- [Monitor your global network with Amazon CloudWatch metrics](#)
- [Monitor your global network using EventBridge](#)
- [Log AWS Global Networks for Transit Gateways API calls using AWS CloudTrail](#)

Monitor your global network with Amazon CloudWatch metrics

You can monitor AWS Global Networks for Transit Gateways using Amazon CloudWatch, which collects raw data and processes it into readable, near-real-time metrics. These statistics are kept for 15 months, so that you can access historical information and gain a better perspective on how your web application or service is performing. You can also set alarms that watch for certain thresholds, and send notifications or take actions when those thresholds are met. For more information, see the [Amazon CloudWatch User Guide](#).

You can view CloudWatch metrics in your global network for your registered transited gateways, your associated Site-to-Site VPN connections, and your on-premises resources. You can view metrics per transit gateway and per transit gateway attachment, per global network.

For more information about the supported metrics, see the following topics:

- [CloudWatch metrics for your transit gateways](#)
- [Monitor VPN tunnels using Amazon CloudWatch](#)
- [View CloudWatch metrics for on-premises resources](#)

For examples of creating alarms, see [Creating Amazon CloudWatch Alarms](#) in the *Amazon CloudWatch User Guide*.

View CloudWatch metrics for on-premises resources

AWS Global Networks for Transit Gateways publishes data points to Amazon CloudWatch for your on-premises resources, including devices and links. CloudWatch enables you to retrieve statistics about those data points as an ordered set of time series data, known as metrics. Each data point has an associated timestamp and an optional unit of measurement.

You can use metrics to verify that your system is performing as expected. For example, you can create a CloudWatch alarm to monitor a specified metric and initiate an action (such as sending a notification to an email address) if the metric goes outside what you consider an acceptable range.

Device metrics

The `AWS/NetworkManager` namespace includes the following metrics for devices.

Metric	Description
<code>DataIn</code>	The number of bytes received by the device.
<code>DataOut</code>	The number of bytes sent by the device.
<code>TunnelDownCount</code>	The number of VPN tunnels on the device that have a DOWN status. Static VPN tunnels with a DOWN status, and BGP VPN tunnels with any state other than ESTABLISHED, are included in the count.

Metric dimensions for devices

To filter the metrics for your devices, use the following dimensions.

Dimension	Description
DeviceId	Filters the metric data by the device.

Link metrics

The AWS/NetworkManager namespace includes the following metrics for links.

Metric	Description
DataIn	The number of bytes received by the on-premises network using this link.
DataOut	The number of bytes sent from the on-premises network using this link.

Metric dimensions for links

To filter the metrics for your links, use the following dimensions.

Dimension	Description
LinkId	Filters the metric data by the link.

View global network CloudWatch metrics

There are various options for viewing CloudWatch metrics for your global network, including the following:

- Viewing metrics for the global network and filtering by transit gateway
- Viewing metrics for a specific transit gateway and its attachments

To view metrics for your global network and filter by transit gateway

1. Open the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.

2. In the navigation pane, choose **Global networks**, and choose the ID for your global network.
3. In the navigation pane, choose **Transit gateway network**.
4. Choose **Monitoring**. On this page, you can filter by transit gateway to view metrics for that transit gateway.

To view metrics for a specific transit gateway and its attachments

1. Open the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. In the navigation pane, choose **Global networks**, and choose the ID for your global network.
3. In the navigation pane, choose **Transit gateways**, and choose the ID for your transit gateway.
4. (Optional) Metrics and events use the default time set up in the CloudWatch Events event. To set a custom time frame, choose **Custom** and then choose a **Relative** or **Absolute** time, and then choose if you want to see that date range in **UTC** or the edge location's **Local time zone**.

Choose **Add to dashboard** to add this metric to your CloudWatch dashboard. For more information about using CloudWatch dashboards, see [Using Amazon CloudWatch Dashboards](#) in the *Amazon CloudWatch User Guide*.

Note

The **Add to dashboard** option only works if your registered transit gateway is in the US West (Oregon) Region.

Monitor your global network using EventBridge

Amazon EventBridge delivers a near-real-time stream of system events that describe changes in your resources. Using simple rules that you can quickly set up, you can match events and route them to one or more target functions or streams. For more information, see the [Amazon EventBridge User Guide](#).

AWS Global Networks for Transit Gateways sends the following types of events to EventBridge:

- [Topology change events](#)
- [Routing update events](#)
- [Status update events](#)

Get started

Before you can view events for your global network, you must onboard to CloudWatch Logs Insights. In the global networks console, choose the ID of your global network. In the **Network events summary** section, choose **Onboard to CloudWatch Log Insights**.

An IAM principal in your account, such as an IAM user, must have sufficient permissions to onboard to CloudWatch Logs Insights. Ensure that the IAM policy contains the following permissions.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "events:PutTargets",
        "events:DescribeRule",
        "logs:PutResourcePolicy",
        "logs:DescribeLogGroups",
        "logs:DescribeResourcePolicies",
        "events:PutRule",
        "logs:CreateLogGroup"
      ],
      "Resource": "*"
    }
  ]
}
```

The preceding policy does not grant permission to create, modify, or delete Network Manager resources. For more information about IAM policies for working with Network Manager, see [Identity and access management for AWS Global Networks for Transit Gateways](#).

When you onboard to CloudWatch Logs Insights, the following occurs:

- A CloudWatch event rule with the name `DON_NOT_DELETE_networkmanager_rule` is created in the US West (Oregon) Region.
- A CloudWatch Logs log group with the name `/aws/events/networkmanagerloggroup` is created in the US West (Oregon) Region.

- The CloudWatch event rule is configured with the CloudWatch Logs log group as a target.
- A CloudWatch resource policy with the name `DO_NOT_DELETE_networkmanager_TrustEventsToStoreLogEvents` is created in the US West (Oregon) Region. To view this policy, use the following AWS CLI command: `aws logs describe-resource-policies --region us-west-2`

View transit gateway events using the AWS Transit Gateway console

You can view events for your global network or view a specific transit gateway using the global networks console.

To view global network events

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateway network**.
5. Choose **Events**.

On this page you can view events for your transit gateway network. For more information about this page, see [the section called “Events”](#).

To view events for a specific transit gateway

1. Access the Network Manager console at <https://console.aws.amazon.com/networkmanager/home/>.
2. Under **Connectivity**, choose **Global Networks**.
3. On the **Global networks** page, choose the global network ID.
4. In the navigation pane, choose **Transit gateways**.
5. Choose the **Transit gateway ID**.
6. Choose **Events**.

On this page you can view events for your transit gateway network. For more information about this page, see [the section called “Events”](#).

Topology change events

Topology change events occur when there have been changes to the resources in your global network. These include the following:

Events

- [A transit gateway in the global network was deleted \(TGW-DELETED\)](#)
- [A VPN connection was created for a transit gateway \(VPN-CONNECTION-CREATED\)](#)
- [A VPN connection was deleted on a transit gateway \(VPN-CONNECTION-DELETED\)](#)
- [The customer gateway for a VPN connection was changed \(VPN-CONNECTION-CUSTOMER-GATEWAY-MODIFIED\)](#)
- [The target gateway for a VPN connection was changed \(VPN-CONNECTION-TARGET-GATEWAY-MODIFIED\)](#)
- [A VPC was attached to a transit gateway \(VPC-ATTACHMENT-CREATED\)](#)
- [A VPC attachment was deleted from a transit gateway \(VPC-ATTACHMENT-DELETED\)](#)
- [An AWS Direct Connect gateway was attached to a transit gateway \(DXGW-ATTACHMENT-CREATED\)](#)
- [An AWS Direct Connect gateway was detached from a transit gateway \(DXGW-ATTACHMENT-DELETED\)](#)
- [A transit gateway peering connection attachment was created \(TGW_PEERING_CREATED\)](#)
- [A transit gateway peering connection was deleted \(TGW-PEERING-DELETED\)](#)
- [A transit gateway Connect attachment was created for a transit gateway \(CONNECT_ATTACHMENT_CREATED\)](#)
- [A transit gateway Connect attachment was deleted for a transit gateway \(CONNECT_ATTACHMENT_DELETED\)](#)
- [A transit gateway Connect peer was created in a Connect attachment \(TGW-CONNECT-PEER-CREATED\)](#)
- [A transit gateway Connect peer was deleted in a Connect attachment \(CONNECT_PEER_DELETED\)](#)
- [A Network Firewall attachment was created \(NETWORK-FIREWALL-ATTACHMENT-CREATED\)](#)
- [A Network Firewall attachment was deleted \(NETWORK-FIREWALL-ATTACHMENT-DELETED\)](#)

A transit gateway in the global network was deleted (TGW-DELETED)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-18T22:18:44Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "TGW-DELETED",
    "changeDescription": "A Transit Gateway in the global network has been deleted.",
    "region": "us-east-1",
    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  }
}
```

A VPN connection was created for a transit gateway (VPN-CONNECTION-CREATED)

```
{
  "version": "0",
  "id": "7636f496-ba9f-b1cc-22a6-8c90bbca8540",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-18T19:52:42Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-01c3c3738ab9f83c5"
  ],
  "detail": {
    "changeType": "VPN-CONNECTION-CREATED",
    "changeDescription": "A Site-to-Site VPN connection has been created.",
    "region": "us-east-1",
  }
}
```

```

    "transitGatewayAttachmentArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "vpnConnectionArn": "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-1234567890abcdef0",
    "customerGatewayArn": "arn:aws:ec2:us-east-1:123456789012:customer-gateway/cgw-1234567890abcdef0",
    "outsideIpAddresses": ["54.166.146.158", "3.93.214.172"],
    "routing": "dynamic_route",
    "accelerated": false,
    "isPrivateVpn": false,
    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  }
}

```

A VPN connection was deleted on a transit gateway (VPN-CONNECTION-DELETED)

```

{
  "version": "0",
  "id": "877fe5fd-4c95-1553-84ef-cfa271121081",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-19T19:43:12Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPN-CONNECTION-DELETED",
    "changeDescription": "A Site-to-Site VPN connection has been deleted.",
    "region": "us-east-1",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "vpnConnectionArn": "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-1234567890abcdef0",
    "customerGatewayArn": "arn:aws:ec2:us-east-1:123456789012:customer-gateway/cgw-1234567890abcdef0",
    "isPrivateVpn": false,
  }
}

```

```

    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/
    tgw-1234567890abcdef0"
  }
}

```

The customer gateway for a VPN connection was changed (VPN-CONNECTION-CUSTOMER-GATEWAY-MODIFIED)

```

{"version": "0",
 "id": "76594f68-2b9f-7885-895e-58ece42ac48a",
 "detail-type": "Network Manager Topology Change",
 "source": "aws.networkmanager",
 "account": "123456789012", "time": "2023-06-28T19:25:12Z",
 "region": "us-west-2",
 "resources": [
   "arn:aws:networkmanager::123456789012:global-network/global-
   network-1234567890abcdef0",
   "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-0822025a9ea3dde43"
 ],
 "detail": {
   "changeType": "VPN-CONNECTION-CUSTOMER-GATEWAY-MODIFIED",
   "changeDescription": "The customer gateway of a Site-to-Site VPN connection has been
   modified",
   "region": "us-east-1",
   "vpnConnectionArn": "arn:aws:ec2:us-east-1:123456789012:vpn-connection/
   vpn-1234567890abcdef0",
   "previousCustomerGatewayArn": "arn:aws:ec2:us-east-1:123456789012:customer-gateway/
   cgw-1234567890abcdef0",
   "currentCustomerGatewayArn": "arn:aws:ec2:us-east-1:123456789012:customer-gateway/
   cgw-1234567890abcdef0",
   "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/
   tgw-1234567890abcdef0"
 }
}

```

The target gateway for a VPN connection was changed (VPN-CONNECTION-TARGET-GATEWAY-MODIFIED)

```

{"version": "0",
 "id": "668a4e46-a757-3663-dc32-308c5ac5d87f",
 "detail-type": "Network Manager Topology Change",

```

```

"source": "aws.networkmanager",
"account": "503089527312",
"time": "2023-06-27T18:27:24Z",
"region": "us-west-2",
"resources": [
  "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
  "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-1234567890abcdef0"
],
"detail": {
  "changeType": "VPN-CONNECTION-TARGET-GATEWAY-MODIFIED",
  "changeDescription": "The target gateway of a Site-to-Site VPN connection has been modified",
  "region": "us-east-1",
  "vpnConnectionArn": "arn:aws:ec2:us-east-1:123456789012:vpn-connection/vpn-1234567890abcdef0",
  "previousTargetGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0",
  "currentTargetGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0",
  "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
}
}

```

A VPC was attached to a transit gateway (VPC-ATTACHMENT-CREATED)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-18T19:52:52Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPC-ATTACHMENT-CREATED",
    "changeDescription": "A VPC attachment has been created for a Core Network.",
  }
}

```

```

    "edgeLocation": "us-east-2",
    "attachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
    "vpcArn": "arn:aws:ec2:us-east-2:123456789012:vpc/vpc-1234567890abcdef0",
    "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  }
}

```

A VPC attachment was deleted from a transit gateway (VPC-ATTACHMENT-DELETED)

```

{
  "account": "123456789012",
  "region": "us-west-2",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "version": "0",
  "time": "2019-06-30T23:18:50Z",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPC-ATTACHMENT-DELETED",
    "changeDescription": "A VPC attachment has been deleted.",
    "region": "us-east-1",
    "transit-gateway-arn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-11111111111122222",
    "transit-gateway-attachment-arn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "vpc-arn": "arn:aws:ec2:us-east-1:123456789012:vpc/vpc-1234567890abcdef0"
  }
}

```

An AWS Direct Connect gateway was attached to a transit gateway (DXGW-ATTACHMENT-CREATED)

```

{
  "version": "0",

```

```

{id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
"detail-type": "Network Manager Topology Change",
"source": "aws.networkmanager",
"account": "123456789012",
"time": "2023-01-19T18:57:29Z",
"region": "us-west-2",
"resources": [
  "arn:aws:networkmanager::123456789012:global-network/global-
network-1234567890abcdef0",
  "arn:aws:ec2:us-west-1:123456789012:transit-gateway/tgw-1234567890abcdef0"],
"detail": {
  "changeType": "DXGW-ATTACHMENT-CREATED",
  "changeDescription": "A Direct Connect Gateway attachment has been created.",
  "region": "us-west-1",
  "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-1:123456789012:transit-
gateway-attachment/tgw-attach-1234567890abcdef0",
  "directConnectGatewayArn": "arn:aws:directconnect::123456789012:dx-gateway/
a1b2c3d4-5678-90ab-cdef-EXAMPLE22222",
  "transitGatewayArn": "arn:aws:ec2:us-west-1:123456789012:transit-gateway/
tgw-1234567890abcdef0"
}
}

```

An AWS Direct Connect gateway was detached from a transit gateway (DXGW-ATTACHMENT-DELETED)

```

{
"version": "0",
"id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
"detail-type": "Network Manager Topology Change",
"source": "aws.networkmanager",
"account": "123456789012",
"time": "2023-01-19T19:16:23Z",
"region": "us-west-2",
"resources": [
  "arn:aws:networkmanager::123456789012:global-network/global-
network-1234567890abcdef0",
  "arn:aws:ec2:us-west-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
],
"detail": {
  "changeType": "DXGW-ATTACHMENT-DELETED",
  "changeDescription": "A Direct Connect Gateway attachment has been deleted.",
  "region": "us-west-1",

```



```

    "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-1:123456789012:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "directConnectGatewayArn": "arn:aws:directconnect::123456789012:dx-gateway/a1b2c3d4-5678-90ab-cdef-EXAMPLE22222",
    "transitGatewayArn": "arn:aws:ec2:us-west-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  }
}

```

A transit gateway peering connection attachment was created (TGW_PEERING_CREATED)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-18T22:28:51Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "TGW_PEERING_CREATED",
    "changeDescription": "A Transit Gateway peering has been created for a Core Network.",
    "edgeLocation": "us-east-1",
    "peeringArn": "arn:aws:networkmanager::123456789012:peering/peering-1234567890abcdef0",
    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0",
    "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  }
}

```

A transit gateway peering connection was deleted (TGW-PEERING-DELETED)

```

{

```

```

"version":"0",
"id":"a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
"detail-type":"Network Manager Topology Change",
"source":"aws.networkmanager",
"account":"503089527312",
"time":"2023-06-27T19:55:59Z",
"region":"us-west-2",
"resources":[
  "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
  "arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
],
"detail":{
  "changeType":"TGW-PEERING-DELETED",
  "changeDescription":"A Transit Gateway peering attachment has been deleted.",
  "region":"us-east-1",
  "transitGatewayAttachmentArn":"arn:aws:ec2:us-east-1:111122223333:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
  "peeredTransitGatewayArn":"arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0",
  "transitGatewayArn":"arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
}
}

```

A transit gateway Connect attachment was created for a transit gateway (CONNECT_ATTACHMENT_CREATED)

```

{
  "version":"0",
  "id":"a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type":"Network Manager Topology Change",
  "source":"aws.networkmanager",
  "account":"123456789012",
  "time":"2022-11-21T23:23:46Z",
  "region":"us-west-2",
  "resources":[
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  ],
  "detail":{
    "changeType":"CONNECT_ATTACHMENT_CREATED",

```

```

    "changeDescription": "A Connect attachment has been created for a Core Network.",
    "edgeLocation": "us-east-1",
    "attachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
    "transportAttachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
    "protocol": "GRE",
    "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  }
}

```

A transit gateway Connect attachment was deleted for a transit gateway (CONNECT_ATTACHMENT_DELETED)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-19T19:26:26Z",
  "region": "us-west-2", "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "CONNECT_ATTACHMENT_DELETED",
    "changeDescription": "A Connect attachment has been deleted for a Core Network.",
    "edgeLocation": "us-east-1",
    "attachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
    "transportAttachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
    "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
  }
}

```

A transit gateway Connect peer was created in a Connect attachment (TGW-CONNECT-PEER-CREATED)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-06-27T17:22:45Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "TGW-CONNECT-PEER-CREATED",
    "changeDescription": "A TGW Connect Peer has been created in a Connect attachment.",
    "region": "us-east-1",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-east-1:111122223333:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "connectPeerArn": "arn:aws:ec2:us-east-1:111122223333:transit-gateway-connect-peer/tgw-connect-peer-1234567890abcdef0",
    "peerAddress": "10.1.2.3",
    "transitGatewayAddress": "10.0.0.1", 111122223333
    "transitGatewayArn": "arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
  }
}
```

A transit gateway Connect peer was deleted in a Connect attachment (CONNECT_PEER_DELETED)

```
{
  "version": "0",
  "id": "437f664b-cc6c-ccb8-b322-2c185ebe0c10",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "738040852526",
  "time": "2023-11-13T20:49:34Z",
```

```

    "region": "us-west-2",
    "resources": [
        "arn:aws:networkmanager::738040852526:global-network/global-
network-02e49afd6fa01d0c3",
        "arn:aws:networkmanager::738040852526:core-network/core-
network-0d6ee69cdc931f7b5"
    ],
    "detail": {
        "changeType": "CONNECT_PEER_DELETED",
        "changeDescription": "A Connect peer has been deleted in a Connect
attachment.",
        "edgeLocation": "eu-west-2",
        "attachmentArn": "arn:aws:networkmanager::738040852526:attachment/
attachment-05e447f0df042a011",
        "connectPeerArn": "arn:aws:networkmanager::738040852526:connect-peer/connect-
peer-024b3172d38112df5",
        "coreNetworkArn": "arn:aws:networkmanager::738040852526:core-network/core-
network-0d6ee69cdc931f7b5"
    }
}

```

A Network Firewall attachment was created (NETWORK-FIREWALL-ATTACHMENT-CREATED)

```

{
    "version": "0",
    "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
    "detail-type": "Network Manager Topology Change",
    "source": "aws.networkmanager",
    "account": "123456789012",
    "time": "2023-01-19T18:57:29Z",
    "region": "us-west-2",
    "resources": [
        "arn:aws:networkmanager::123456789012:global-network/global-
network-1234567890abcdef0",
        "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
    ],
    "detail": {
        "changeType": " ",
        "changeDescription": "A Network Firewall attachment has been created.",
        "region": "us-east-1",
        "transitGatewayAttachmentArn": "arn:aws:ec2:us-east-1:123456789012:transit-
gateway-attachment/tgw-attach-1234567890abcdef0",
    }
}

```

```

    "resourceArns":[
      "arn:aws:network-firewall:us-east-1:123456789012:firewall/arn:aws:network-
firewall:us-east-1:123456789012:firewall/firewall-network-manager-event-example"
    ],
    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/
tgw-1234567890abcdef0"
  }
}

```

A Network Firewall attachment was deleted (NETWORK-FIREWALL-ATTACHMENT-DELETED)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-19T19:16:23Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-
network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:123456789012:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "NETWORK-FIREWALL-ATTACHMENT-DELETED",
    "changeDescription": "A Network Firewall attachment has been deleted.",
    "region": "us-east-1",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-east-1:123456789012:transit-
gateway-attachment/tgw-attach-1234567890abcdef0",
    "resourceArns":[
      "arn:aws:network-firewall:us-east-1:123456789012:firewall/arn:aws:network-
firewall:us-east-1:123456789012:firewall/firewall-network-manager-event-example"
    ],
    "transitGatewayArn": "arn:aws:ec2:us-east-1:123456789012:transit-gateway/
tgw-1234567890abcdef0"
  }
}

```

Routing update events

Routing update events occur when there have been changes to the transit gateway route tables in your global network. These include the following:

Events

- [A transit gateway attachment's route table changed \(CONNECT_PEER_DELETED\)](#)
- [A route was created in a transit gateway route table \(TGW-ROUTE-INSTALLED\)](#)
- [A route was deleted in a transit gateway route table gateway \(TGW-ROUTE-UNINSTALLED\)](#)

A transit gateway attachment's route table changed (CONNECT_PEER_DELETED)

```
{
  "version": "0",
  "id": "437f664b-cc6c-ccb8-b322-2c185ebe0c10",
  "detail-type": "Network Manager Topology Change",
  "source": "aws.networkmanager",
  "account": "738040852526",
  "time": "2023-11-13T20:49:34Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::738040852526:global-network/global-network-02e49afd6fa01d0c3",
    "arn:aws:networkmanager::738040852526:core-network/core-network-0d6ee69cdc931f7b5"
  ],
  "detail": {
    "changeType": "CONNECT_PEER_DELETED",
    "changeDescription": "A Connect peer has been deleted in a Connect attachment.",
    "edgeLocation": "eu-west-2",
    "attachmentArn": "arn:aws:networkmanager::738040852526:attachment/attachment-05e447f0df042a011",
    "connectPeerArn": "arn:aws:networkmanager::738040852526:connect-peer/connect-peer-024b3172d38112df5",
    "coreNetworkArn": "arn:aws:networkmanager::738040852526:core-network/core-network-0d6ee69cdc931f7b5"
  }
}
```

A route was created in a transit gateway route table (TGW-ROUTE-INSTALLED)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Routing Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-06-27T15:24:32Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "TGW-ROUTE-INSTALLED",
    "changeDescription": "Routes in one or more Transit Gateway route tables have been installed.",
    "region": "us-east-1",
    "transitGatewayRouteTableArns": [
      "arn:aws:ec2:us-east-1:111122223333:transit-gateway-route-table/tgw-rtb-1234567890abcdef0"
    ],
    "sequenceNumber": 1687879467281,
    "routes": [{
      "destinationCidrBlock": "11.0.0.0/16",
      "attachments": [
        {
          "tgwAttachmentId": "tgw-attach-1234567890abcdef0",
          "resourceId": "vpc-1234567890abcdef0",
          "attachmentType": "vpc"
        }
      ],
      "routeType": {
        "route_propagated",
        "routeState": "active",
        "propagatedRouteFamily": {
          "connected" }
      },
      "transitGatewayArn": "arn:aws:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
    }
  ]
}
```


A route was deleted in a transit gateway route table gateway (TGW-ROUTE-UNINSTALLED)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Routing Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2022-02-30T23:18:50Z",
  "region": "us-east-1",
  "resources": [
    "arn:aws-us-east-1:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws-us-east-1:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "TGW-ROUTE-UNINSTALLED",
    "changeDescription": "Routes in one or more Transit Gateway route tables have been uninstalled.",
    "region": "us-east-1",
    "transitGatewayRouteTableArns": [
      "arn:aws-us-east-1:ec2:us-east-1:111122223333:transit-gateway-route-table/tgw-rtb-1234567890abcdef0"
    ],
    "sequenceNumber": 1648147298451,
    "routes": [{
      "destinationCidrBlock": "10.10.10.0/16",
      "attachments": [],
      "routeType": "route_static",
      "routeState": "blackhole"
    }
  ],
  "transitGatewayArn": "arn:aws-us-east-1:ec2:us-east-1:111122223333:transit-gateway/tgw-1234567890abcdef0"
}
```

Status update events

Status update events occur when there have been changes to the status of the connectivity of your VPN connections in the global network. These include the following:

Events

- [A VPN tunnel's IPsec session went down \(VPN-CONNECTION-IPSEC-DOWN\)](#)
- [A VPN tunnel's IPsec session went up \(after being down\) \(VPN-CONNECTION-IPSEC-UP\)](#)
- [A VPN tunnel's BGP session went down \(VPN-CONNECTION-BGP-DOWN\)](#)
- [A VPN tunnel's BGP session went up \(after being down\) \(VPN-CONNECTION-BGP-ESTABLISH\)](#)
- [A Connect peer \(GRE tunnel\) BGP session went down \(CONNECT_PEER_BGP_DOWN\)](#)
- [A Connect peer \(GRE tunnel\) BGP session went up after being down \(CONNECT_PEER_BGP_UP\)](#)

A VPN tunnel's IPsec session went down (VPN-CONNECTION-IPSEC-DOWN)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Status Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-31T19:48:05Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPN-CONNECTION-IPSEC-DOWN",
    "changeDescription": "IPsec for a VPN connection has gone down.",
    "region": "us-west-2",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "vpnConnectionArn": "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-1234567890abcdef0",
    "outsideIpAddress": "35.84.102.207",
    "transitGatewayArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway/tgw-1234567890abcdef0"
  }
}
```

```
}
}
```

A VPN tunnel's IPsec session went up (after being down) (VPN-CONNECTION-IPSEC-UP)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Status Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-31T19:34:54Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-network-1234567890abcdef0",
    "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPN-CONNECTION-IPSEC-UP",
    "changeDescription": "IPsec for a VPN connection has come up.",
    "region": "us-west-2",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway-attachment/tgw-attach-1234567890abcdef0",
    "vpnConnectionArn": "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-1234567890abcdef0",
    "outsideIpAddress": "52.37.214.193",
    "transitGatewayArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway/tgw-1234567890abcdef0"
  }
}
```

A VPN tunnel's BGP session went down (VPN-CONNECTION-BGP-DOWN)

```
{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Status Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-31T19:48:23Z",
```

```

    "region": "us-west-2",
    "resources": [
      "arn:aws:networkmanager::123456789012:global-network/global-
network-0c243052669618f74",
      "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-0fdb136628eff65a8"
    ],
    "detail": {
      "changeType": "VPN-CONNECTION-BGP-DOWN",
      "changeDescription": "BGP for a VPN connection has gone down.",
      "region": "us-west-2",
      "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-2:111122223333:transit-
gateway-attachment/tgw-attach-1234567890abcdef0",
      "vpnConnectionArn": "arn:aws:ec2:us-west-2:111122223333:vpn-connection/
vpn-1234567890abcdef0",
      "outsideIpAddress": "54.190.210.71",
      "peerAsn": "65001",
      "transitGatewayArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway/
tgw-1234567890abcdef0"
    }
  }
}

```

A VPN tunnel's BGP session went up (after being down) (VPN-CONNECTION-BGP-ESTABLISH)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Status Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-01-31T19:34:40Z",
  "region": "us-west-2",
  "resources": [
    "arn:aws:networkmanager::123456789012:global-network/global-
network-1234567890abcdef0",
    "arn:aws:ec2:us-west-2:111122223333:vpn-connection/vpn-1234567890abcdef0"
  ],
  "detail": {
    "changeType": "VPN-CONNECTION-BGP-ESTABLISH",
    "changeDescription": "BGP for a VPN connection has been established.",
    "region": "us-west-2",
    "transitGatewayAttachmentArn": "arn:aws:ec2:us-west-2:111122223333:transit-
gateway-attachment/tgw-attach-1234567890abcdef0",

```

```

    "vpnConnectionArn": "arn:aws:ec2:us-west-2:111122223333:vpn-connection/
vpn-1234567890abcdef0",
    "outsideIpAddress": "52.37.214.193",
    "peerAsn": "65001",
    "transitGatewayArn": "arn:aws:ec2:us-west-2:111122223333:transit-gateway/
tgw-1234567890abcdef0"
  }
}

```

A Connect peer (GRE tunnel) BGP session went down (CONNECT_PEER_BGP_DOWN)

```

{
  "version": "0",
  "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
  "detail-type": "Network Manager Status Update",
  "source": "aws.networkmanager",
  "account": "123456789012",
  "time": "2023-03-01T19:57:34Z",
  "region": "us-west-2",
  "resources": ["arn:aws:networkmanager::123456789012:global-network/global-
network-07a82dd610af0cc57", "arn:aws:networkmanager::123456789012:core-network/core-
network-1234567890abcdef0"],
  "detail": {
    "changeType": "CONNECT_PEER_BGP_DOWN",
    "changeDescription": "BGP for a Connect peer has gone down.",
    "edgeLocation": "ap-southeast-1",
    "attachmentArn": "arn:aws:networkmanager::123456789012:attachment/
attachment-1234567890abcdef0",
    "connectPeerArn": "arn:aws:networkmanager::123456789012:connect-peer/connect-
peer-1234567890abcdef0",
    "peerAsn": "65011",
    "coreNetworkAddress": "192.0.2.0",
    "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-
network-1234567890abcdef0"
  }
}

```

A Connect peer (GRE tunnel) BGP session went up after being down (CONNECT_PEER_BGP_UP)

```

{

```

```

    "version": "0",
    "id": "a1b2c3d4-5678-90ab-cdef-EXAMPLE11111",
    "detail-type": "Network Manager Status Update",
    "source": "aws.networkmanager",
    "account": "123456789012",
    "time": "2023-03-01T19:57:49Z",
    "region": "us-west-2",
    "resources": ["arn:aws:networkmanager::123456789012:global-network/global-network-07a82dd610af0cc57", "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"],
    "detail": {
      "changeType": "CONNECT_PEER_BGP_UP",
      "changeDescription": "BGP for a Connect peer has been established.",
      "edgeLocation": "ap-southeast-1",
      "attachmentArn": "arn:aws:networkmanager::123456789012:attachment/attachment-1234567890abcdef0",
      "connectPeerArn": "arn:aws:networkmanager::123456789012:connect-peer/connect-peer-1234567890abcdef0",
      "peerAsn": "65011",
      "coreNetworkAddress": "192.0.2.0",
      "coreNetworkArn": "arn:aws:networkmanager::123456789012:core-network/core-network-1234567890abcdef0"
    }
  }
}

```

Log AWS Global Networks for Transit Gateways API calls using AWS CloudTrail

AWS Global Networks for Transit Gateways works with AWS CloudTrail, a service that provides a record of actions taken by a user, role, or an AWS service in global networks. CloudTrail captures all API calls for global network as events. The calls that are captured include calls from the Network Manager console and code calls to the global API operations. If you create a trail, you can enable continuous delivery of CloudTrail events to an Amazon S3 bucket, including events for Global Networks. If you don't configure a trail, you can still view the most recent events in the CloudTrail console in **Event history**. Using the information collected by CloudTrail, you can determine what request was made to global networks, the IP address from which the request was made, who made the request, when it was made, and additional details.

To learn more about CloudTrail, see the [AWS CloudTrail User Guide](#).

Global network information in CloudTrail

CloudTrail is enabled on your AWS account when you create the account. When activity occurs in a global network, that activity is recorded in a CloudTrail event along with other AWS service events in **Event history**. You can view, search, and download recent events in your AWS account. For more information, see [Viewing Events with CloudTrail Event History](#).

For an ongoing record of events in your AWS account, including events for a global network create a trail. A *trail* enables CloudTrail to deliver log files to an Amazon S3 bucket. By default, when you create a trail in the console, the trail applies to all AWS Regions. The trail logs events from all Regions in the AWS partition, and delivers the log files to the Amazon S3 bucket that you specify. Additionally, you can configure other AWS services to further analyze and act upon the event data collected in CloudTrail logs. For more information, see the following topics in the *AWS CloudTrail User Guide*.

- [Overview for Creating a Trail](#)
- [CloudTrail Supported Services and Integrations](#)
- [Receiving CloudTrail Log Files from Multiple Regions](#) and [Receiving CloudTrail Log Files from Multiple Accounts](#)

All actions in a global network are logged by CloudTrail and are documented in the [Network Manager API Reference](#). For example, calls to the `CreateGlobalNetwork` action generate entries in the CloudTrail log files.

Every event or log entry contains information about who generated the request. The identity information helps you determine the following:

- Whether the request was made with root or AWS Identity and Access Management (IAM) user credentials
- Whether the request was made with temporary security credentials for a role or federated user
- Whether the request was made by another AWS service

For more information, see the [CloudTrail userIdentity Element](#).

Identity and access management for AWS Global Networks for Transit Gateways

AWS Identity and Access Management (IAM) is an AWS service that helps an administrator securely control access to AWS resources. IAM administrators control who can be *authenticated* (signed in) and *authorized* (have permissions) to use AWS Global Networks for Transit Gateways resources. IAM is an AWS service that you can use with no additional charge. You can use features of IAM to allow other users, services, and applications to use your AWS resources fully or in a limited way, without sharing your security credentials.

By default, IAM users don't have permission to create, view, or modify AWS resources. To allow an IAM user to access resources, such as a global network, and perform tasks, you must:

- Create an IAM policy that grants the IAM user permission to use the specific resources and API actions they need
- Attach the policy to the IAM user or to the group to which the IAM user belongs

When you attach a policy to a user or group of users, it allows or denies the user permissions to perform the specified tasks on the specified resources.

Important

If you grant access to a global network in Network Manager, you grant access to all AWS service data associated with the registered transit gateways across all Regions.

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How Network Manager works with IAM

With IAM identity-based policies, you can specify allowed or denied actions and resources, and specify the conditions under which actions are allowed or denied. Network Manager supports specific actions, resources, and condition keys. For a complete list, see [Actions, Resources, and Condition Keys for AWS Network Manager](#) in the *Service Authorization Reference*.

To learn about all of the elements that you use in a JSON policy, see [IAM JSON Policy Elements Reference](#) in the *IAM User Guide*.

Actions

Policy actions in Network Manager use the following prefix before the action: `networkmanager:`. For example, to grant someone permission to create a global network with the `CreateGlobalNetwork` API operation, you include the `networkmanager:CreateGlobalNetwork` action in their policy.

For a list of global networks actions, see the [Network Manager API Reference](#).

Resources

The Resource element specifies the object or objects to which the action applies. Statements must include either a Resource or a NotResource element. You specify a resource using an ARN or using the wildcard (*) to indicate that the statement applies to all resources.

The global network resource has the following ARN.

```
arn:${Partition}:networkmanager:${Account}:global-network/${GlobalNetworkId}
```

For example, to specify the `global-network-1122334455aabbccd` global network in your statement, use the following ARN.

```
"Resource": "arn:aws:networkmanager::123456789012:global-network/global-network-1122334455aabbccd"
```

Condition keys

The Condition element (or Condition *block*) lets you specify conditions in which a statement is in effect. The Condition element is optional. You can build conditional expressions that use

[condition operators](#), such as equals or less than, to match the condition in the policy with values in the request.

If you specify multiple Condition elements in a statement, or multiple keys in a single Condition element, AWS evaluates them using a logical AND operation. If you specify multiple values for a single condition key, AWS evaluates the condition using a logical OR operation. All of the conditions must be met before the statement's permissions are granted.

You can also use placeholder variables when you specify conditions. For example, you can grant an IAM user permission to access a resource only if it is tagged with their IAM user name. For more information, see [IAM Policy Elements: Variables and Tags](#) in the *IAM User Guide*.

You can attach tags to global networks resources or pass tags in a request to global networks. To control access based on tags, you provide tag information in the [condition element](#) of a policy using the `aws:ResourceTag/key-name`, `aws:RequestTag/key-name`, or `aws:TagKeys` condition keys.

To see all AWS global condition keys, see [AWS Global Condition Context Keys](#) in the *IAM User Guide*.

global networks also supports the following condition keys:

- `networkmanager:tgwArn`—Controls which transit gateways can be registered or deregistered in your global network.
- `networkmanager:cgwArn`—Controls which customer gateways can be associated or disassociated from devices and links in your global network.
- `networkmanager:tgwConnectPeerArn`—Controls which Connect peers can be associated or disassociated from devices and links in your global network.

Example policies to manage global networks

The following are example IAM policies for working with global networks.

Administrator access

The following IAM policy grants full access to the Amazon EC2, global networks, AWS Direct Connect, and CloudWatch APIs. This enables administrators to create and manage transit gateways and their attachments (such as VPCs and AWS Direct Connect gateways), create and manage global

networks resources, and monitor global networks using CloudWatch metrics and events. The policy also grants user permissions to create any required service-linked roles.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "networkmanager:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "cloudwatch:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "events:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "directconnect:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "iam:CreateServiceLinkedRole",
      "Resource": "arn:aws:iam::*:role/aws-service-role/*"
    }
  ]
}
```

Read-only access

The following IAM policy grants read-only access to the Amazon EC2, global networks, AWS Direct Connect, CloudWatch, and EventBridge APIs. This enables users to use the global networks console to view and monitor global networks and their associated resources, and view metrics and events for the resources. Users cannot create or modify any resources.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:Get*",
        "ec2:Describe*"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "networkmanager:Get*",
        "networkmanager:Describe*"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "cloudwatch:List*",
        "cloudwatch:Get*",
        "cloudwatch:Describe*"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "logs:Describe*",
        "logs:Get*",

```

```

        "logs:List*",
        "logs:StartQuery",
        "logs:StopQuery",
        "logs:TestMetricFilter",
        "logs:FilterLogEvents"
    ],
    "Resource": "*"
},
{
    "Effect": "Allow",
    "Action": [
        "events:List*",
        "events:TestEventPattern",
        "events:Describe*"
    ],
    "Resource": "*"
},
{
    "Effect": "Allow",
    "Action": "directconnect:Describe*",
    "Resource": "*"
}
]
}

```

JSON

```

{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "networkmanager:CreateSite"
            ],
            "Resource": [
                "arn:aws:networkmanager::111122223333:global-network/global-network-1122334455aabbccd",
                "arn:aws:networkmanager::111122223333:site/*"
            ]
        },
        {

```

```

        "Effect": "Allow",
        "Action": [
            "networkmanager:CreateDevice"
        ],
        "Resource": [
            "arn:aws:networkmanager::111122223333:global-network/global-network-1122334455aabbccd",
            "arn:aws:networkmanager::111122223333:site/*",
            "arn:aws:networkmanager::111122223333:device/*"
        ]
    },
    {
        "Effect": "Allow",
        "Action": [
            "networkmanager:CreateLink"
        ],
        "Resource": [
            "arn:aws:networkmanager::111122223333:global-network/global-network-1122334455aabbccd",
            "arn:aws:networkmanager::111122223333:link/*"
        ]
    }
]
}

```

Controlling the use of transit gateways and customer gateways

The following IAM policy enables users to work with global networks resources, but they are explicitly denied permission to do the following:

- Register or deregister a specific transit gateway (tgw-aabbccdd112233445) in the global network.
- Associate or disassociate a specific customer gateway (cgw-11223344556677abc) in the global network.

The policy uses the `networkmanager:tgwArn` and `networkmanager:cgwArn` condition keys to enforce these conditions.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "networkmanager:*"
      ],
      "Resource": [
        "*"
      ]
    },
    {
      "Effect": "Deny",
      "Action": [
        "networkmanager:RegisterTransitGateway",
        "networkmanager:DeregisterTransitGateway"
      ],
      "Resource": [
        "*"
      ],
      "Condition": {
        "StringEquals": {
          "networkmanager:tgwArn": "arn:aws:ec2:region:account-
id:transit-gateway/tgw-aabbccdd112233445"
        }
      }
    },
    {
      "Effect": "Deny",
      "Action": [
        "networkmanager:AssociateCustomerGateway",
        "networkmanager:DisassociateCustomerGateway"
      ],
      "Resource": [
        "*"
      ],
      "Condition": {
        "StringEquals": {
          "networkmanager:cgwArn": "arn:aws:ec2:region:account-
id:customer-gateway/cgw-11223344556677abc"
        }
      }
    }
  ]
}
```

```
}  
  }  
    }  
      ]  
        }
```

AWS Global Networks for Transit Gateways service-linked roles

AWS Global Networks for Transit Gateways uses service-linked roles for the permissions that it requires to call other AWS services on your behalf. These service-linked roles are not propagated to your AWS Organizations management account.

Permissions granted by the service-linked role

AWS Global Networks for Transit Gateways uses a Network Manager service-linked role named `AWSServiceRoleForNetworkManager` to call the actions on your behalf when you work with global networks.

The `AWSServiceRoleForNetworkManager` service-linked role trusts the following service to assume the role:

- `networkmanager.amazonaws.com`

This service-linked role uses the managed policy `AWSNetworkManagerServiceRolePolicy`. To view the permissions for this policy, see [AWSNetworkManagerServiceRolePolicy](#) in the *AWS Managed Policy Reference*.

Create the service-linked role

You don't need to manually create the `AWSServiceRoleForNetworkManager` role. global networks creates this role for you when you create your first global network.

For global networks to create a service-linked role on your behalf, you must have the required permissions. For more information, see [Service-linked role permissions](#) in the *IAM User Guide*.

Edit the service-linked role

You can edit the description of **AWSServiceRoleForNetworkManager** using IAM. For more information, see [Edit a service-linked role description](#) in the *IAM User Guide*.

Delete the service-linked role

If you no longer need to use global networks, we recommend that you delete the **AWSServiceRoleForNetworkManager** role.

You can delete this service-linked role only after you delete your global network. For information about how to delete your global network, see [Delete a global network](#).

You can use the IAM console, the IAM CLI, or the IAM API to delete service-linked roles. For more information, see [Delete a service-linked role](#) in the *IAM User Guide*.

After you delete **AWSServiceRoleForNetworkManager**, Network Manager will create the role again when you create a new global network.

Supported Regions for AWS Global Networks for Transit Gateways service-linked roles

AWS Global Networks for Transit Gateways supports the custom-linked roles in all of AWS Regions where the service is available. For more information, see [the section called "Region availability"](#).

AWS managed policies for AWS Global Networks for Transit Gateways

To add permissions to users, groups, and roles, it is easier to use AWS managed policies than to write policies yourself. It takes time and expertise to [create IAM customer managed policies](#) that provide your team with only the permissions they need. To get started quickly, you can use our AWS managed policies. These policies cover common use cases and are available in your AWS account. For more information about AWS managed policies, see [AWS managed policies](#) in the *IAM User Guide*.

AWS services maintain and update AWS managed policies. You can't change the permissions in AWS managed policies. Services occasionally add additional permissions to an AWS managed

policy to support new features. This type of update affects all identities (users, groups, and roles) where the policy is attached. Services are most likely to update an AWS managed policy when a new feature is launched or when new operations become available. Services do not remove permissions from an AWS managed policy, so policy updates won't break your existing permissions.

Additionally, AWS supports managed policies for job functions that span multiple services. For example, the `ReadOnlyAccess` AWS managed policy provides read-only access to all AWS services and resources. When a service launches a new feature, AWS adds read-only permissions for new operations and resources. For a list and descriptions of job function policies, see [AWS managed policies for job functions](#) in the *IAM User Guide*.

AWS managed policy: `NetworkAdministrator`

You can attach the `NetworkAdministrator` policy to your IAM identities. This policy grants permissions that allow registered delegated administrators and the management account *administrator* access to global networks. For more information, see [the section called "Multi-account access roles"](#).

To view the permissions for this policy, see [NetworkAdministrator](#) in the *AWS Managed Policy Reference*.

AWS managed policy: `AWSNetworkManagerReadOnlyAccess`

You can attach the `AWSNetworkManagerReadOnlyAccess` policy to your IAM identities. This policy grants permissions that allow registered delegated administrators and the management account *read-only* access to global networks. For more information, see [the section called "Multi-account access roles"](#).

To view the permissions for this policy, see [AWSNetworkManagerReadOnlyAccess](#) in the *AWS Managed Policy Reference*.

AWS managed policy: `AWSNetworkManagerServiceRolePolicy`

This policy is attached to the service-linked role named `AWSServiceRoleForNetworkManager` to allow AWS Global Networks for Transit Gateways to call API actions on your behalf when you work with global networks. For more information, see [the section called "Service-linked role"](#).

To view the permissions for this policy, see [AWSNetworkManagerServiceRolePolicy](#) in the *AWS Managed Policy Reference*.

AWS Global Networks for Transit Gateways updates to AWS managed policies

View details about updates to AWS managed policies for Network Manager since this service began tracking these changes in April 2021. For automatic alerts about changes to this page, subscribe to the RSS feed on the Network Manager Document history page.

Change	Description	Date
AWSNetworkManagerServiceRolePolicy	AWS Global Networks for Transit Gateways added permission to call the following API action: <code>GetTransitGatewayRouteTablePropagations</code> .	July 12, 2022
NetworkAdministrator	AWS Global Networks for Transit Gateways began using administrative permissions in member accounts for multi-account access.	May 24, 2022
AWSNetworkManagerReadOnlyAccess - Updated existing policy	AWS Global Networks for Transit Gateways began using read-only permissions in member accounts for multi-account access.	May 24, 2022
AWSNetworkManagerServiceRolePolicy - Updated existing policy	AWS Global Networks for Transit Gateways added permission to call the following API actions: <code>organizations:DescribeAccount</code> , <code>organizations:DescribeOrganization</code> ,	May 24, 2022

Change	Description	Date
	organizations:ListAccounts , organizations:ListAWSServiceAccessForOrganization , and organizations:ListDelegatedAdministrators .	
AWSNetworkManagerServiceRolePolicy - Updated existing policy	AWS Global Networks for Transit Gateways added permissions to call the following API action: ec2:DescribeRegions .	December 2, 2021
AWSNetworkManagerServiceRolePolicy : updated existing policy	AWS Global Networks for Transit Gateways added permissions to call the following API actions: directconnect:DescribeDirectConnectGateways , ec2:DescribeVpnConnections , ec2:DescribeVpcs , ec2:GetTransitGatewayRouteTableAssociations , ec2:SearchTransitGatewayRoutes , ec2:DescribeTransitGatewayPeeringAttachments , ec2:DescribeTransitGatewayConnects , and ec2:DescribeTransitGatewayConnectPeers .	June 1, 2021

Multi-account access roles for AWS Global Networks for Transit Gateways

AWS Global Networks for Transit Gateways uses AWS CloudFormation StackSets to deploy and manage the following two custom IAM roles in AWS Organizations member accounts to support multi-account permissions. These two roles are deployed to every member account in the organization when AWSServiceAccess is enabled (trusted access). For more information about multi-account, see [Manage multiple accounts in global networks using AWS Organizations](#).

The custom IAM roles are created automatically through the Network Manager service when you enable multi-account access using the global networks console. We strongly recommend that you use the console for enabling multi-account. Choosing an alternative approach requires an advanced level of expertise, and opens the multi-account for your global network to be more prone to error.

CloudWatch-CrossAccountSharingRole

This policy provides delegated administrators and the management accounts access to CloudWatch monitoring data from other member accounts. The following is an example of the template.

```
AWSTemplateFormatVersion: '2010-09-09'
Description: Enables CloudWatch in central monitoring accounts to assume permissions to
  view CloudWatch data in the current account

Resources:
  CloudWatch-CrossAccountSharingRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: CloudWatch-CrossAccountSharingRole
      AssumeRolePolicyDocument:
        Version: '2012-10-17'
        Statement:
          - Effect: Allow
            Principal:
              AWS: [
                "arn:aws:iam::<account1-id>:root",
                "arn:aws:iam::<account2-id>:root",
                "arn:aws:iam::<account3-id>:root"
              ]
            Action:
              - sts:AssumeRole
```

```

Path: "/"
ManagedPolicyArns:
  - arn:aws:iam::aws:policy/CloudWatchReadOnlyAccess

```

IAMRoleForAWSNetworkManagerCrossAccountResourceAccess

The `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess` IAM policy role, based on your selection when enabling trusted access through the global networks console, enables either administrative or read-only global networks console switch role access. An associated administrative or read-only template is also deployed along with the policy. For information about these templates, see [the section called "Permission templates"](#).

The following is an example of the administrator role template.

```

AWSTemplateFormatVersion: '2010-09-09'
Description: Enables admin cross account resource access through switch role

Resources:
  IAMRoleForAWSNetworkManagerCrossAccountResourceAccess:
    Type: AWS::IAM::Role
    Properties:
      RoleName: IAMRoleForAWSNetworkManagerCrossAccountResourceAccess
      AssumeRolePolicyDocument:
        Version: '2012-10-17'
        Statement:
          - Effect: Allow
            Principal:
              AWS: [
                "arn:aws:iam::<account1-id>:root",
                "arn:aws:iam::<account2-id>:root",
                "arn:aws:iam::<account3-id>:root"
              ]
            Action:
              - sts:AssumeRole
      Path: "/"
      ManagedPolicyArns:
        - arn:aws:iam::aws:policy/NetworkAdministrator

```

The following is the read-only role template.

```

AWSTemplateFormatVersion: '2010-09-09'

```

Description: Enables read only cross account resource access through switch role

Resources:

`IAMRoleForAWSNetworkManagerCrossAccountResourceAccess:`

Type: `AWS::IAM::Role`

Properties:

RoleName: `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess`

AssumeRolePolicyDocument:

Version: `'2012-10-17'`

Statement:

- Effect: `Allow`

Principal:

```

AWS: [
    "arn:aws:iam::<account1-id>:root",
    "arn:aws:iam::<account2-id>:root",
    "arn:aws:iam::<account3-id>:root"
]

```

Action:

- `sts:AssumeRole`

Path: `"/"`

ManagedPolicyArns:

- `arn:aws:iam::aws:policy/AWSNetworkManagerReadOnlyAccess`
- `arn:aws:iam::aws:policy/AWSDirectConnectReadOnlyAccess`
- `arn:aws:iam::aws:policy/AmazonEC2ReadOnlyAccess`
- `arn:aws:iam::aws:policy/AmazonRoute53ReadOnlyAccess`

Permission templates

When choosing the `IAMRoleForAWSNetworkManagerCrossAccountResourceAccess` permission, an associated administrative or read-only template is also passed to AWS CloudFormation StackSets. These templates contain a list of accounts that are able to assume these roles. These accounts include the AWS Organizations management account and all registered delegated administrators for the Network Manager service. Deregistering a delegated administrator removes it from this list so that it can no longer assume these roles. Disabling trusted access deletes the AWS CloudFormation StackSets, and in turn all member account stacks and custom IAM roles in those accounts that were StackSets-managed for multi-account.

NetworkAdministrator

This policy enables administrator permission for the delegated administrator and management accounts to modify resources from other accounts in the global network while using the Network Manager console switch role.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "autoscaling:Describe*",
        "cloudfront:ListDistributions",
        "cloudwatch:DeleteAlarms",
        "cloudwatch:DescribeAlarms",
        "cloudwatch:GetMetricStatistics",
        "cloudwatch:PutMetricAlarm",
        "directconnect:*",
        "ec2:AcceptVpcEndpointConnections",
        "ec2:AllocateAddress",
        "ec2:AssignIpv6Addresses",
        "ec2:AssignPrivateIpAddresses",
        "ec2:AssociateAddress",
        "ec2:AssociateDhcpOptions",
        "ec2:AssociateRouteTable",
        "ec2:AssociateSubnetCidrBlock",
        "ec2:AssociateVpcCidrBlock",
        "ec2:AttachInternetGateway",
        "ec2:AttachNetworkInterface",
        "ec2:AttachVpnGateway",
        "ec2:CreateCarrierGateway",
        "ec2:CreateCustomerGateway",
        "ec2:CreateDefaultSubnet",
        "ec2:CreateDefaultVpc",
        "ec2:CreateDhcpOptions",
        "ec2:CreateEgressOnlyInternetGateway",
        "ec2:CreateFlowLogs",
        "ec2:CreateInternetGateway",
        "ec2:CreateNatGateway",
        "ec2:CreateNetworkAcl",
        "ec2:CreateNetworkAclEntry",
        "ec2:CreateNetworkInterface",
        "ec2:CreateNetworkInterfacePermission",
        "ec2:CreatePlacementGroup",
        "ec2:CreateRoute",
        "ec2:CreateRouteTable",

```



```
"ec2:CreateSecurityGroup",
"ec2:CreateSubnet",
"ec2:CreateTags",
"ec2:CreateVpc",
"ec2:CreateVpcEndpoint",
"ec2:CreateVpcEndpointConnectionNotification",
"ec2:CreateVpcEndpointServiceConfiguration",
"ec2:CreateVpnConnection",
"ec2:CreateVpnConnectionRoute",
"ec2:CreateVpnGateway",
"ec2:DeleteCarrierGateway",
"ec2:DeleteEgressOnlyInternetGateway",
"ec2:DeleteFlowLogs",
"ec2:DeleteNatGateway",
"ec2:DeleteNetworkInterface",
"ec2:DeleteNetworkInterfacePermission",
"ec2:DeletePlacementGroup",
"ec2:DeleteSubnet",
"ec2:DeleteTags",
"ec2:DeleteVpc",
"ec2:DeleteVpcEndpointConnectionNotifications",
"ec2:DeleteVpcEndpointServiceConfigurations",
"ec2:DeleteVpcEndpoints",
"ec2:DeleteVpnConnection",
"ec2:DeleteVpnConnectionRoute",
"ec2:DeleteVpnGateway",
"ec2:DescribeAccountAttributes",
"ec2:DescribeAddresses",
"ec2:DescribeAvailabilityZones",
"ec2:DescribeCarrierGateways",
"ec2:DescribeClassicLinkInstances",
"ec2:DescribeCustomerGateways",
"ec2:DescribeDhcpOptions",
"ec2:DescribeEgressOnlyInternetGateways",
"ec2:DescribeFlowLogs",
"ec2:DescribeInstances",
"ec2:DescribeInternetGateways",
"ec2:DescribeKeyPairs",
"ec2:DescribeMovingAddresses",
"ec2:DescribeNatGateways",
"ec2:DescribeNetworkAcls",
"ec2:DescribeNetworkInterfaceAttribute",
"ec2:DescribeNetworkInterfacePermissions",
"ec2:DescribeNetworkInterfaces",
```

```
"ec2:DescribePlacementGroups",
"ec2:DescribePrefixLists",
"ec2:DescribeRouteTables",
"ec2:DescribeSecurityGroupReferences",
"ec2:DescribeSecurityGroupRules",
"ec2:DescribeSecurityGroups",
"ec2:DescribeStaleSecurityGroups",
"ec2:DescribeSubnets",
"ec2:DescribeTags",
"ec2:DescribeVpcAttribute",
"ec2:DescribeVpcClassicLink",
"ec2:DescribeVpcClassicLinkDnsSupport",
"ec2:DescribeVpcEndpointConnectionNotifications",
"ec2:DescribeVpcEndpointConnections",
"ec2:DescribeVpcEndpointServiceConfigurations",
"ec2:DescribeVpcEndpointServicePermissions",
"ec2:DescribeVpcEndpointServices",
"ec2:DescribeVpcEndpoints",
"ec2:DescribeVpcPeeringConnections",
"ec2:DescribeVpcs",
"ec2:DescribeVpnConnections",
"ec2:DescribeVpnGateways",
"ec2:DescribePublicIpv4Pools",
"ec2:DescribeIpv6Pools",
"ec2:DetachInternetGateway",
"ec2:DetachNetworkInterface",
"ec2:DetachVpnGateway",
"ec2:DisableVgwRoutePropagation",
"ec2:DisableVpcClassicLinkDnsSupport",
"ec2:DisassociateAddress",
"ec2:DisassociateRouteTable",
"ec2:DisassociateSubnetCidrBlock",
"ec2:DisassociateVpcCidrBlock",
"ec2:EnableVgwRoutePropagation",
"ec2:EnableVpcClassicLinkDnsSupport",
"ec2:ModifyNetworkInterfaceAttribute",
"ec2:ModifySecurityGroupRules",
"ec2:ModifySubnetAttribute",
"ec2:ModifyVpcAttribute",
"ec2:ModifyVpcEndpoint",
"ec2:ModifyVpcEndpointConnectionNotification",
"ec2:ModifyVpcEndpointServiceConfiguration",
"ec2:ModifyVpcEndpointServicePermissions",
"ec2:ModifyVpcPeeringConnectionOptions",
```

```

        "ec2:ModifyVpcTenancy",
        "ec2:MoveAddressToVpc",
        "ec2:RejectVpcEndpointConnections",
        "ec2:ReleaseAddress",
        "ec2:ReplaceNetworkAclAssociation",
        "ec2:ReplaceNetworkAclEntry",
        "ec2:ReplaceRoute",
        "ec2:ReplaceRouteTableAssociation",
        "ec2:ResetNetworkInterfaceAttribute",
        "ec2:RestoreAddressToClassic",
        "ec2:UnassignIpv6Addresses",
        "ec2:UnassignPrivateIpAddresses",
        "ec2:UpdateSecurityGroupRuleDescriptionsEgress",
        "ec2:UpdateSecurityGroupRuleDescriptionsIngress",
        "elasticbeanstalk:Describe*",
        "elasticbeanstalk:List*",
        "elasticbeanstalk:RequestEnvironmentInfo",
        "elasticbeanstalk:RetrieveEnvironmentInfo",
        "elasticloadbalancing:*",
        "logs:DescribeLogGroups",
        "logs:DescribeLogStreams",
        "logs:GetLogEvents",
        "route53:*",
        "route53domains:*",
        "sns:CreateTopic",
        "sns:ListSubscriptionsByTopic",
        "sns:ListTopics"
    ],
    "Resource": "*"
},
{
    "Effect": "Allow",
    "Action": [
        "ec2:AcceptVpcPeeringConnection",
        "ec2:AttachClassicLinkVpc",
        "ec2:AuthorizeSecurityGroupEgress",
        "ec2:AuthorizeSecurityGroupIngress",
        "ec2:CreateVpcPeeringConnection",
        "ec2>DeleteCustomerGateway",
        "ec2>DeleteDhcpOptions",
        "ec2>DeleteInternetGateway",
        "ec2>DeleteNetworkAcl",
        "ec2>DeleteNetworkAclEntry",
        "ec2>DeleteRoute",

```

```

        "ec2:DeleteRouteTable",
        "ec2:DeleteSecurityGroup",
        "ec2:DeleteVolume",
        "ec2:DeleteVpcPeeringConnection",
        "ec2:DetachClassicLinkVpc",
        "ec2:DisableVpcClassicLink",
        "ec2:EnableVpcClassicLink",
        "ec2:GetConsoleScreenshot",
        "ec2:RejectVpcPeeringConnection",
        "ec2:RevokeSecurityGroupEgress",
        "ec2:RevokeSecurityGroupIngress"
    ],
    "Resource": [
        "*"
    ]
},
{
    "Effect": "Allow",
    "Action": [
        "ec2:CreateLocalGatewayRoute",
        "ec2:CreateLocalGatewayRouteTableVpcAssociation",
        "ec2:DeleteLocalGatewayRoute",
        "ec2:DeleteLocalGatewayRouteTableVpcAssociation",

"ec2:DescribeLocalGatewayRouteTableVirtualInterfaceGroupAssociations",
        "ec2:DescribeLocalGatewayRouteTableVpcAssociations",
        "ec2:DescribeLocalGatewayRouteTables",
        "ec2:DescribeLocalGatewayVirtualInterfaceGroups",
        "ec2:DescribeLocalGatewayVirtualInterfaces",
        "ec2:DescribeLocalGateways",
        "ec2:SearchLocalGatewayRoutes"
    ],
    "Resource": "*"
},
{
    "Effect": "Allow",
    "Action": [
        "s3:GetBucketLocation",
        "s3:GetBucketWebsite",
        "s3:ListBucket"
    ],
    "Resource": [
        "*"
    ]
}

```

```

    },
    {
      "Effect": "Allow",
      "Action": [
        "iam:GetRole",
        "iam:ListRoles",
        "iam:PassRole"
      ],
      "Resource": "arn:aws:iam::*:role/flow-logs-*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "networkmanager:*"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "ec2:AcceptTransitGatewayVpcAttachment",
        "ec2:AssociateTransitGatewayRouteTable",
        "ec2:CreateTransitGateway",
        "ec2:CreateTransitGatewayRoute",
        "ec2:CreateTransitGatewayRouteTable",
        "ec2:CreateTransitGatewayVpcAttachment",
        "ec2>DeleteTransitGateway",
        "ec2>DeleteTransitGatewayRoute",
        "ec2>DeleteTransitGatewayRouteTable",
        "ec2>DeleteTransitGatewayVpcAttachment",
        "ec2:DescribeTransitGatewayAttachments",
        "ec2:DescribeTransitGatewayRouteTables",
        "ec2:DescribeTransitGatewayVpcAttachments",
        "ec2:DescribeTransitGateways",
        "ec2:DisableTransitGatewayRouteTablePropagation",
        "ec2:DisassociateTransitGatewayRouteTable",
        "ec2:EnableTransitGatewayRouteTablePropagation",
        "ec2:ExportTransitGatewayRoutes",
        "ec2:GetTransitGatewayAttachmentPropagations",
        "ec2:GetTransitGatewayRouteTableAssociations",
        "ec2:GetTransitGatewayRouteTablePropagations",
        "ec2:ModifyTransitGateway",
        "ec2:ModifyTransitGatewayVpcAttachment",
        "ec2:RejectTransitGatewayVpcAttachment",

```

```

        "ec2:ReplaceTransitGatewayRoute",
        "ec2:SearchTransitGatewayRoutes"
    ],
    "Resource": "*"
  },
  {
    "Effect": "Allow",
    "Action": [
      "iam:CreateServiceLinkedRole"
    ],
    "Resource": "*",
    "Condition": {
      "StringLike": {
        "iam:AWSServiceName": [
          "transitgateway.amazonaws.com"
        ]
      }
    }
  }
]
}

```

AWSNetworkManagerReadOnlyAccess

This policy enables read-only permission for the delegated administrator and management accounts to review information about resources from other accounts in the global network while using the global networks console switch role, but doesn't allow either account to make changes.

JSON

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "networkmanager:Describe*",
        "networkmanager:Get*",
        "networkmanager:List*"
      ],
      "Resource": "*"
    }
  ]
}

```

```
}  
  ]  
}
```

AWS Global Networks for Transit Gateways Quotas

Your AWS account has the quotas shown in the following table for AWS Global Networks for Transit Gateways.

The Service Quotas console also provides information about global networks quotas. You can use the Service Quotas console to view default quotas and [request quota increases](#) for adjustable quotas. For more information, see [Requesting a quota increase](#) in the *Service Quotas User Guide*.

General quotas

The following global networks general quotas apply.

Quota	Default	Adjustable
Global networks per AWS account	5	Yes
Number of devices per global network	200	Yes
Number of sites per global network	200	Yes
Number of links per global network	200	Yes
Number of connections per global network	200	Yes
Number of registered delegated administrators for an organization in AWS Organizations	10	Yes

Document history for AWS Global Networks for Transit Gateways

Change	Description	Date
Support for dual-stack IPv6 endpoints.	Network Manager now supports dual-stack IPv6 endpoints.	March 25, 2025
Support PrivateLink.	Network Manager now supports PrivateLink in the us-west-2 and us-gov-west-1 Regions.	March 25, 2025
Support PrivateLink.	Network Manager now supports PrivateLink in the us-west-2 and us-gov-west-1 Regions.	March 25, 2025
Support PrivateLink.	Network Manager now supports PrivateLink in the us-west-2 and us-gov-west-1 Regions.	March 25, 2025
New Region support	Added ap-southeast-5 to the list of available Regions.	March 13, 2025
New Region support	Added me-central-1 , ap-south-2 , ap-southeast-4 , ca-west-1 , eu-south-2 , and eu-central-2 to the list of available Regions.	October 3, 2024
New Region support	Added ap-southeast-3 to the list of available Regions.	March 26, 2024

Guide renamed	The <i>Network Manager User Guide</i> was renamed to the <i>AWS Global Networks for Transit Gateways User Guide</i> , as the AWS Network Manager console was reorganized. AWS Global Networks for Transit Gateways are now a part of the greater-feature Network Manager console.	November 28, 2022
Updated AWS managed policy	Network Manager added a new action, <code>GetTransitGatewayRouteTablePropagations</code> , to its AWS managed policies.	July 12, 2022
Multi-account support	Network Manager now supports multi-account, which allows you to centrally manage multiple AWS Organizations accounts and transit gateways in a single global network.	May 24, 2022
New guide created	Network Manager documentation was removed from the <i>AWS Transit Gateway User Guide</i> and included as part of a new, standalone <i>AWS Network Manager User Guide</i> .	December 2, 2021
Documentation updated for AWS Cloud WAN	The <i>AWS Network Manager User Guide</i> was updated, as Network Manager supports both AWS Transit Gateways and AWS Cloud WAN.	December 2, 2021